CHAPTER I

Buddhism

1 Introduction

Epistemology, the theory of knowledge, is a philosophical branch that examines the nature, origins, limitations, and validity of knowledge. The term "epistemology" derives from the Greek terms "episteme" and "logos". The term "logos" can be interpreted as "account," "argue," or "reason," whereas "episteme" may be comprehended as "knowledge", "understanding", or "acquaintance". Each of these varied translations encapsulates a distinct aspect of the meaning of the Greek phrase while simultaneously reflecting multiple dimensions of epistemology. Although the term "epistemology" is relatively recent, the discipline itself is as ancient as any other philosophical branch. Throughout the extensive history of epistemology, many periods have garnered significant attention in specific parts of the field. Plato's epistemology aimed to elucidate the essence of knowledge, distinguishing it from mere accurate opinion, while emphasising its significance for the individual who holds it. Kant's epistemology sought to examine the conditions that facilitate human understanding, while Russell's epistemology aimed to substantiate modern science through sensory experience. Locke's epistemology aims to elucidate the processes of human knowledge, whereas Kant's endeavours to investigate the prerequisites that facilitate human understanding. Recent endeavours in formal epistemology have focused on clarifying how our confidence levels are rationally influenced by our evidence. Conversely, modern feminist epistemology examines how our desires influence our evidence and, more broadly, our rational limitations.

Indian philosophy

Indian philosophy has a longer history of continuous development than any other philosophical tradition, featuring a wide variety of schools and systems. Virtually every Indian philosophical tradition is associated with a religious sect, including the six orthodox (astika) schools (darshanas) such as Nyaya, Vaisheshika, Samkhya, Yoga, Purva Mimamsa, and Vedanta, alongside the heterodox (nastika) schools, which consist of Buddhism, Jainism, and the materialistic Carvaka school. All Indian philosophical traditions demonstrate numerous shared ideas. In the Indian context, discussions within educational institutions primarily centre on epistemological issues; however, no author examines knowledge devoid of the metaphysical

commitments intrinsic to various classical systems (darśana), encompassing realist and idealist, dualist and monist, as well as theist and atheist viewpoints, among others. Each of the principal schools, totalling approximately twelve, delineates a position on knowledge and justification from the inception of its history, even if, as demonstrated by the Buddhist sceptic (Prasaṅgika), it predominantly challenges the theories of others. Nevertheless, certain common epistemological assumptions or attitudes are present, the most significant of which is a focus on the origin of belief or justification. Mainstream classical Indian epistemology is predominantly defined by notions of genealogy, particularly with knowledge-generating processes known as pramāṇas, which translates to "knowledge sources". The principal options are perception, inference, and testimony. Other processes seem to lack truth-conduciveness or cannot be distilled from one or more of the widely recognised sources, such as perception and inference.

Since antiquity, the essence of knowledge, referred to as 'jñāna', has been a central focus in philosophy. This subject has attracted the most attention from intellectuals globally. This concept holds significance throughout various systems of Indian philosophy. Every philosophy posits that ignorance is the principal cause of human suffering. Consequently, possessing precise knowledge is vital to eliminate pain from the world. What precisely constitutes knowledge? What are the sources of the information? What is the criterion for assessing the veracity of knowledge? In summary, these are several epistemological issues that drive the formulation of a theory of knowledge within the domain of epistemology. Epistemology, a branch of philosophy, examines the nature of human knowledge and the means of its justification. This branch of philosophy examines the origin, nature, techniques, validity, and limits of knowing.

The notion of knowledge can be regarded as a conviction about reality within the framework of philosophical examination. All practical endeavours are based on knowledge. The objective of knowledge is to illuminate what is inherently obscure. A known object and a conscious person are both essential elements of knowledge. Knowledge cannot exist without both a subject and an object of knowledge. The individual possesses this property. Several alternative terms for knowledge include buddhi, denoting understanding; upalabdhi, signifying apprehension; and pratyaya, representing concepts, judgement, awareness, and cognition. The definitions of these two terms are synonymous. In Indian epistemology, the notion of knowing is denoted by two terms: jñāna and pramā.

The name 'jñāna' is derived from the root 'jna', meaning 'to know'. All forms of knowledge, including correct, incorrect, valid, invalid, true, and false, belong to the domain of Jñāna.

'Pramā' is a term formed from the root 'ma' and the prefix 'pra', signifying valid knowledge. According to Nagesa, valid knowledge is entirely novel and aligns with its object (avisaṁvadijñāna). The term 'pramā' is solely utilised to denote authentic knowledge, referred to as 'yathārtha jñāna'. It is utilised to signify knowledge from a psychological standpoint that aids in comprehending an object's true nature. Knowledge was employed not only for metaphysics during the Upaniṣadic era but also for several other objectives. The term 'jñāna' in Advaita Vedanta denotes both absolute knowledge (svarūpa jñāna) and empirical knowledge (vṛtti jñāna). They contend that vṛtti jñāna qualifies as 'jñāna' when illuminated by the principle of consciousness, referred to as Brahman (or atman). Various schools of thought exist within Indian philosophy about the concept of knowing. This essay will analyse the perspectives and knowledge of only two schools: Nyāya and Buddhism.

Buddhism

Buddhist epistemology is the philosophical examination of knowledge and legitimate cognition within the Buddhist tradition. A main focus is the process of acquiring authentic knowledge and how this knowledge facilitates the eradication of ignorance (avijjā), the fundamental root of suffering. Buddhist epistemology, unlike most other Indian philosophical traditions, is distinctly factual, analytical, and focused on liberation. It prioritises actual experience and rational analysis over metaphysical conjecture.

Each philosophical tradition in India has endeavoured to formulate a theory of knowledge, referred to as the Pramana Sastra, that underpins the metaphysical structure of the respective school. Although the primary objective of philosophy is widely recognised as the acquisition of knowledge of reality (tattva-jhana), it is thought that a theory of knowledge is an essential prerequisite for a theory of reality. The premise that philosophising involves contemplating the essence of reality as revealed through experience underpins this belief. On the other hand, because every experience involves a cognitive reference to an object, there is always a persistent risk of misinterpreting this connection. Although every experience contains an inherent trans phenomenality or self-transcendence, it does not guarantee that it will reveal its object in an acceptable and truthful manner. An inquiry about the veracity of the experience is necessary due to the potential for error therein. This work serves as the foundation and center of the entire epistemological endeavour. Consequently, scholars within the Buddhist tradition strive to construct a robust epistemological basis for their understanding of reality.

Buddhist philosophy is a moral framework that emerged in India between the sixth and fifth centuries BCE. It is a non-theistic tradition of Indian philosophy. Initially, the sagacious Siddhartha Gautama founded it. During a time when India was undergoing a significant religious and intellectual transformation, the Buddha established a belief system that would subsequently be recognised as Christianity. Upon attaining this realisation, the Buddha discerned that the origins of suffering were desire and attachment and that humanity's suffering stemmed from ignorance of the true nature of existence.

Buddhist philosophy has focused on a no-substance ontology. The profound desire to eliminate suffering drove Gautama, the Buddha, who founded Buddhist philosophy, to contemplation. He conceptualised that all pain arises from 'tanha' (desire) and that all desire stems from connection to the illusory belief in permanence. He sought to propose a solution to the labyrinth of suffering, presenting impermanence or nonsubstantiality as the essential means to transcend suffering. The Four Noble Truths support the concepts of impermanence and nonsubstantiality as means to eliminate suffering.

The Buddha has interpreted the entirety of reality, typically perceived through the lenses of matter and consciousness, as a succession of transient moments. The term "moment" implies that all reality or existence is intrinsically linked to time. Everything that exists is confined within the limits of time. Consequently, an existence series is synonymous with a time series. The existence series may be physical, psychological, or a synthesis of both. Each series is, in essence, a conglomerate. A material entity is a physical agglomeration (bhuta sanghdta), while a mental entity is a psychical aggregation (citia sanghdta). In Buddhism, wisdom is characterised by understanding and discernment. It includes self-awareness of the body and mind, reflection on the inevitability of death, and the recognition that all complex entities are subject to change and, hence, are not worthy of pursuit. Wisdom entails acknowledging that humans lack total control over their bodies, thoughts, or personalities; hence, these elements do not define one's inner self and are not possessed by any single individual. Conversely, everything that exists is really an impersonal endeavour. Due to the complexities that govern every event, it is impossible for anyone to exert perfect control over all factors, which means that while we may strive for control, we must also accept the unpredictability of outcomes. Nonetheless, the potential to alter certain life circumstances, such as improving personal skills or achieving specific goals, renders discipline and practice worthwhile, irrespective of their inherent difficulties.

Buddhists were tasked with formulating a theory that elucidates the deceptive nature of standard knowledge acquisition methods and provides a framework for rectifying the faults arising from their use. A notable advancement in this area is the development of arguments contesting the idea that specific practices typically employed to establish beliefs generate knowledge.

An object is defined as something that exists inside the realm of awareness. It is exhibited either directly or indirectly in manifestation. The entity that is directly accessible to consciousness is termed the object of perception, while the entity that is not immediately accessible is termed the object of inference. Conversely, the object of perception is inherently situated within the dimensions of space and time, but the object of inference is initially conceptualised through a sign and then located in a more abstract realm, such as through reasoning or deduction based on available evidence. Consequently, there exist only two categories of objects: objects of perception and objects of inference.

The reliability of knowledge relies on two sources: sensory experience and inference. Only information sourced from these two sources can be accepted as true. The phrase "attainable item" refers to accurate knowledge obtained by sensory perception and inference. Consequently, knowledge acquired through means apart from perception and inference lacks validity. The presence of this information is perpetually an illusion. An essential feature of possessing accurate knowledge is the ability to identify factors that motivate individuals to undertake intentional actions in pursuit of fulfilling their needs. Consequently, knowledge that is both precise and functional is characterised as information that identifies an entity that meets our criteria.

According to several Buddhist schools, "right knowledge is knowledge that is not refuted by experience." Knowledge is accurate and valid when it can refer to a specific entity. Knowledge does not generate or furnish any item; it only directs our focus towards the object. Consequently, knowledge is deemed accurate when it motivates us to pursue attainable objectives and when it directs us toward the truth. False cognition is a type of cognitive process that leads us to perceive an object that differs from the one we possess. An object can assume various shapes, contingent upon diverse spatiotemporal contexts. Consequently, any sort of cognition that discloses merely a singular aspect or shape of an object cannot be regarded as genuine knowledge. Cognition cannot be deemed accurate if it reveals the erroneous position and time of an object.

Dharmakirti

Dharmakirti not only learned the doctrines of Vasubandhu and Dinnaga but also excelled in their application. Seven works by Dharmakirti are recognised and accessible. Among them, Pramana Vartika is the most distinguished. Pramana Vartika represents an enhancement of Dinnaga's perspective articulated in Pramana Samuccaya, and it exceeds the latter work. It comprises four chapters. The first deals with the examination of the pramanas in general, the second with pratyaksa as pramana, the third with inference (svarthanumana), and the fourth with syllogism (pardrthanumana). Dharmakirti's subsequent work is Pramana Viniscaya. It is a summary of Pramana Vartika. The third piece is Nyaya Bindu, which constitutes a further condensation. The remaining four volumes are concise treatises focused on particular subjects. Hetu Bindu is a concise categorisation of logical justifications. Sambandha Pariksd is an analysis of relational issues. Codana Prakarana is a discourse on the practice of engaging in debate. Santanantara Siddhih is a discourse on the nature of other minds. It seeks to disprove solipsism.

“All effective human endeavours are invariably preceded by knowledge.” 2 In this introductory statement, Dharmakirti delineates the scope and objectives of epistemology and logic in the Nyaya Bindu. Human behaviour can be either intentional or instinctive. The aim of humanity is once again either favourable or unfavourable. An intentional action informed by knowledge involves achieving the desirable and evading the unpleasant. Knowledge is effective in facilitating successful action by enabling the achievement of desirable objectives and the avoidance of unwanted outcomes. A cause may be either productive (karaka) or informational (jhdpaka). In that context, knowledge is solely a catalyst for successful behaviour. It enables us to achieve tangible results, which have practical efficacy.

False cognition is distinct from knowledge. Erroneous cognition leads us to pursue an object different from the one identified. Objects vary due to their shape and spatiotemporal positions. Cognition, being a representation of an item in one form, should not be deemed accurate when the actual object possesses a different shape. Similarly, cognition is inaccurate if it misrepresents the location and characteristics of an object.

In the Pranmana Vartika, Dharmakirti defines knowledge as cognition that is congruent with its object (Pramanam avisamvadi jnanam). Three, he asserts that cognition that completely aligns with its object will also exhibit novelty (Ajnata artha prakaso va). Four: It reveals an object that is not yet recognised, as the object is ephemeral, and only knowledge acquired at that precise instant aligns with the object's existence. It is the initial instance of cognition, the moment of primary awareness. Buddhists assert that continuous cognition is not genuine or true, as it constitutes recognition rather than novel cognition. Among these two properties, the

first, absence of discordance, is fundamental and might be considered the distinguishing trait of knowledge. Buddhist philosophy asserts that all phenomena are perpetually in flux. All entities are ephemeral by nature. Dharmakīrti asserts that continuous knowledge is unattainable. He contends that continuous cognition, being merely recognition, lacks authenticity and validity. Continuous cognition is not a novel concept. Consequently, he asserts that knowledge is a type of cognition that reveals the true nature of an object. The object must be disclosed in its authentic state through understanding.

Dignága

Dignaga is a well-known post-Nágárjuna Buddhist philosopher who built on the ideas of Maitreya, Asanga, and Vasubandhu by rejecting Nagarjuna's negative dialectics and replacing them with a philosophical positivism that combines the Sautrantika and Vijnanavada schools. Dignaga is credited for reconciling Sautrantika ontology with Vijnanavada ontology, formulating an epistemology and logic appropriate for this new ontology, and promoting a unique philosophical methodology grounded in the technique of apoha (double negation). He presents a novel approach to philosophy by integrating ontological discourse within an epistemological framework, a method that subsequently captivated the Navya Nyaya philosopher Ganges'a and his adherents. Dignaga employs the method of double negation (apoha) to enhance clarity and precision in cognition and language. He is an enigmatic logician. He promotes a binary classification in his reasoning. This contradiction is manifested in the promotion of two forms of reality, two ways of cognition, and two nuances of meaning. He employs the concept of svatovyavartana (mutual exclusion) in the domains of reality, knowledge, and language. In essence, each real number represents a distinct class, while all other entities form its complementary class. Consequently, there exists a distinct hierarchy within reality, comprising solely two categories of the real: svalaksana (unique specificity) and samanyalaksana (its generalised representation). What is uniquely specific cannot serve as a generalised representation, and vice versa. This inflexible dichotomy is applied to the domain of epistemology. Dinnaga's observation that every epistemology must be organised with consideration for ontological requirements was groundbreaking. In accordance with this ontological commitment in Dignaga’s epistemology, there exist exclusively two valid means of knowing: pratyaksa (perception) and anumana (inference), as well as two categories of knowledge: perceptual and inferential. The perception can only recognise the unique specific and can never comprehend its generalised representation. The inferential, conversely, perceives just the generalised image and is incapable of recognising the unique special. Similarly to the

rigid dichotomy and stringent hierarchy present in reality, there exists a rigid dichotomy and rigorous hierarchy in the realm of knowledge. Dinnaga extends this binary dichotomy to the realm of language and meaning. Each concept can be articulated in language according to its significance. A word signifies a certain concept. The definition of a term is the antithesis of its negation. Consequently, there can be no overlap or intersection in the meanings of any two words.

Dharmottara

Proper knowledge is delineated by Dharmottara, a prominent Buddhist philosopher and the author of an authoritative commentary on the Nyayābindu. He characterises proper knowledge as knowledge that can be validated. In other words, information is accurate when it aligns with objective reality. The objective of knowledge is fully realised when it accurately reflects the core of reality.

Knowledge represents intellect that accurately reflects reality. It signifies that in knowledge, the object must be understood as it truly is, not as something other. A nondeviating reference is a fundamental prerequisite for knowledge; thus, the veracity of knowledge lies in its alignment with the object of cognition. Dharmottara states, "In everyday life, when we assert that truth is being articulated, we imply that it enables us to attain an object." Likewise, cognition is valid because it directs us toward the object it signifies. Knowledge does not generate an object nor present it to us; rather, it enables us to get it. Reaching an object simply means paying attention to it. Five Dharmottara identifies three different, sequential stages in the process of object apprehension, with each subsequent step arising from the prior one. They are perceiving (adhigati), engaging (pravartana), and attaining (prapana). He asserts that the initial stage solely encompasses knowledge.

There are two categories of knowledge. It is instinctive when it originates from inside. It is discursive when it is attained by focusing our attention on an item through the senses and cognitive consciousness. Epistemology exclusively examines discursive knowledge. According to Dharmottra, there exist two categories of knowledge deemed accurate: instinctive and discursive. Knowledge that arises from inside is termed 'instinctual knowledge'. The act of focusing our attention on an object through sensory input exemplifies the presence of discursive knowledge. He asserts that when successfully gaining proper knowledge, we must ensure we have previously encountered it, as our memory motivates our will. Will propels action, and activity culminates in the attainment of the objective. This exemplifies an indirect cause,

defined as one that lacks any intermediate causal chain. On the other hand, Buddhist logic cannot analyse a deliberate action that directly achieves its objectives (instinctive knowledge). Consequently, Dharmottra underscores the need for discursive knowledge. Classical Buddhist epistemologists assert that knowledge episodes can be categorised into two fundamental types: perceptual knowledge and inferential knowledge. Buddhist epistemologists, starting with Dignāga and continuing thereafter, have concentrated on demonstrating that these two Pramanas are the sole possibilities, arguing that all forms of knowledge can ultimately be traced back to either direct perception or logical inference.

In Indian philosophy, pramāṇa signifies a valid means of knowledge—the method by which precise understanding is achieved. Buddhist epistemology offers a distinctive interpretation of pramāṇa, grounded in the Buddha's emphasis on experience, impermanence, and non-self. Buddhism, unlike most Indian philosophical traditions that acknowledge several pramāṇas, adopts a minimalist and factually based approach to knowledge. Most Buddhist traditions, especially those expressed by philosophers such as Dignāga and Dharmakīrti, recognise two pramāṇas: perception (pratyakṣa) and inference (anumāna). These two are considered sufficient for obtaining both secular information and liberating comprehension, as they provide a foundational framework for understanding reality and achieving enlightenment in Buddhist philosophy.

Perception

In Buddhist philosophy, perception is crucial for understanding the development of human experience and the origin of suffering. It is an active cognitive process that detects, interprets, and categorises experiences rather than a mere passive reception of sensory information. The Buddha examined perception to illustrate that our understanding of reality is significantly shaped by mental conditioning, making it fundamentally inaccurate, as our perceptions are often influenced by past experiences, cultural contexts, and personal biases. In Buddhism, perception is termed 'Saññā' in Pāli and 'Saṃjñā' in Sanskrit. It signifies the cognitive ability

of recognition or identification. When a human perceives an object through the senses—such as a shape through sight or a sound through hearing—perception identifies the object by identifying its characteristics and assigning it a concept or label. For example, when perceiving a particular shape and hue, recognition categorises it as "a tree". Thus, perception acts as a bridge between unprocessed sensory data and conceptual understanding.

In Buddhist epistemology, perception (pratyaksa) is considered the primary pramana. In Buddhist tradition, the term 'pramana' denotes both the process of cognition and the information obtained via it. Buddhists do not recognise a separation between the process of knowledge (pramana) and its result (pramanaphala)—prama. Buddhist philosophers assert that a distinction between the noetic process and its result is unattainable. The process of cognising is entirely synonymous with the cognition of an item. For Buddhists, knowledge is self-evident, therefore eliminating the necessity to differentiate between pramana and its phala.

In Buddhist epistemology, perception (pratyaksa) is considered the primary pramana, as inference (anumana) is contingent upon it. Inference relies on perception, so perception holds precedence over inference in this context. While all Buddhist philosophers from Dinnaga onwards agree on this matter, their interpretations of the essence of perception differ, resulting in varied definitions.

In Buddhism, all knowledge is classified as either direct or indirect. The direct is distinct from the indirect, and the indirect is distinct from the direct. Direct cognition is formally referred to as 'pratyaksa', while indirect cognition is termed 'kalpana' or 'paroksa'. Pratyaksa can thus be delineated by its differentiation from kalpana or paroksa. Consequently, pratyaksa can be interpreted as the "negation of kalpana".

In Buddhist epistemology, perception is considered the paramount Pramana. According to Dharmakirti, perception is characterised as the acknowledgement of an object's unique attributes (svalaksana), devoid of any comparison to other entities, irrespective of their similarities or differences, and entirely independent of nomenclature and verbal expressions, which are merely conceptual constructs. The reality that is immediately presented to awareness constitutes the object of perception. This contrasts with elements expressed as thoughts and words (vikalpas), which are not part of immediate sensory input but are instead constructed in the mind, highlighting the distinction between direct experience and conceptual understanding in Buddhist philosophy. Although all Buddhist philosophers, starting with Dignaga and extending thereafter, concur on this subject, their interpretations of perception's essence differ,

resulting in diverse definitions. The object of vision is distinct from both intellectual and verbal representations, as well as from anything that is solely imagined. Svalaksana, the unique attribute, is the only authentic item that is both real and functional, making it the sole element capable of eliciting a sensory response. A figment of imagination is defined as something that is conceived mentally and hence lacks the capacity to produce any form of stimulus.

According to Dharmakirti, perception entails the recognition of an object in its unique essence (svalaksana), devoid of any relation to other similar or dissimilar objects, and entirely independent of names and verbal expressions, which are mere conceptual constructs. The object of perception is the actual entity that is directly revealed to consciousness, as opposed to objects represented as concepts and words (vikalpas), which are not part of immediate sensory input but are instead formulated in the mind.

The object of perception differs not just from conceptual and verbal representations but also from mere figments of imagination. The unique specific (svalaksana) is the sole real object, and as the genuine and effective entity, it alone may elicit a sensation upon the senses. A phantasm of imagination is, by definition, unreal and thus incapable of exerting any such influence. In response to Dharmakirti's assertion in the Nydya Bindu that "all successful human action is preceded by right knowledge," Dharmottara, within the framework of perception, states that when an object is perceived through direct experience, it is transformed into an object of intentional action via sensory perception.

Inference

Anumana is regarded by Buddhists as a legitimate means of knowledge (pramana), alongside pratyaksa. The two distinct categories of knowledge are pratyaksa, signifying "instant", and paroksa, denoting "mediate". Unlike pratyaksa, the approach dubbed 'anumana' results in the attainment of just intermediary (paroksa) knowledge. Similarly, Anumana can be perceived as both an epistemological approach and a mode of reasoning. This results in the presence of both an epistemic and a logical dimension, which are closely interconnected, indicating that understanding Anumana requires both a grasp of its knowledge-based aspects and its reasoning processes. Anumana is generally recognised as a way of knowledge, as the former serves as the foundation for the latter.

Anumana is simultaneously a method of knowledge and a form of reasoning. Consequently, it possesses both an epistemic and a logical dimension, which are inextricably intertwined. The former aspect underlies the latter, which is why anumana is typically recognised as a style of knowledge.

The Buddhist concept of anumana has evolved within the framework of the dual pramanas, specifically pratyaksa and anumana, which is predicated on the notion of two categories of knowing objects (prameyas). The Buddhists propose a process-orientated perspective of reality, contrasting with the substantival approach upheld by other Indian philosophical groups. A tangible entity, characterised by discrete and transient existence, is solely provided through our sensory experience, which consists of pure feelings free from any conceptualisations.

Dinnaga provided a novel direction and energy to the exploration of logic and epistemology within the Buddhist tradition. He deserves commendation for addressing the theory of pramanas independently from the notion of prameyas. In him, and possibly in Vasubandhu preceding him, we observe an endeavour to compose autonomous treatises on pramanas without incorporating specific examinations of prameyas. He asserts that there are solely two pramanas: pratyaksa and anumana. The other pramanas, such as sabda, were diminished to anumana.

For Dinnaga, anumana may comprise a simple propositional cognitive process or may be expressed in a sentential form. A purely propositional anumana is a cognitive process that remains unarticulated verbally. He designated it as svarthdnumana. When the propositions of an anumana are articulated in language (prakasana or akhydna), they assume a sentential (s'abda) quality, and this type of anumana is referred to as pardrthdnumdna. Keith14 asserts that the differentiation between the two types of anumana was initially presented by Dinnaga in Indian logic.

Dharmakirti discusses the concepts in his works, Pramana Vartika and Nyaya Bindu. The latter study distinctly investigates and defines the fundamental concepts and terminology related to the philosophy of anumana. The subsequent sections thoroughly describe Dharmakirti's principal contribution, which is his explication of the doctrine of trairupya. The doctrine of trairupya considers hetu or linga as the foundational element of anumana and establishes its validity based on three requisite requirements that a valid hetu must satisfy. In summary, they are as follows:
Buddhism adopts a three-member syllogism, consisting of:

1. Thesis (pratijñā) – the statement to be proven

2. Reason (hetu) – the logical ground
3. Example (dṛṣṭānta) – an illustration of the universal relation

This simplified structure reflects the Buddhist emphasis on cognitive efficiency and practical reasoning.

1. There should be a necessary presence of hetu in paksa (anumeye sattvarri). This corresponds to the idea of paksadharmatva available in the Nyaya position.
2. There should be a presence of hetu only in such cases that are similar to paksa (sapakseva sattvam).
3. Lastly, there should be a necessary absence of hetu in such cases that are dissimilar to paksa (asapkse casattvameva niscitam).

The second and third conditions pertain to the two categories of vyâpti: affirmative (anvaya) and negative (vyatireki). While Dinnaga introduced these three requirements of hetu or linga, Dharmakirti provided their systematic formulations, which further refined the understanding of inference in Indian philosophy. Indeed, the entire discourse about anumana revolves around its vital role.

Although it appears that Vasubandhu initially established systematic and logical analysis, it was Dinnaga and Dharmakirti who formulated a thorough idea of anumana. Although referenced in the Pitakas, the term "anumana" is not cited within the framework of a scientific profession grounded in logic. The Anumana Sutta establishes the principle that one should abstain from treating others in a way one would not desire for oneself. The generalisation that others share an individual's aversions purports to underpin this idea. This knowledge is described as inferential.

This paradigm reflects the Buddhist focus on cognitive efficiency and practical thinking, having been streamlined accordingly. In the Buddhist tradition, inference serves not only as a means of intellectual exploration but also fulfils a soteriological purpose. Impermanence (anicca), suffering (dukkha), and non-self (anattā) are notions that can be understood by inference prior to persons directly experiencing them via meditation practice. The mind is thereby primed for insight via the process of inference, which diminishes ignorance and erroneous beliefs. Nonetheless, Buddhism recognises the limitations inherent in inference. It is

posited that the sole means of grasping ultimate reality is through direct, non-conceptual perception, as inferential knowledge is conceptual and mediated by cognition. Consequently, inference functions more as a guide than as a definitive realisation.

Knock your brain

1In Indian philosophy, the sources of valid knowledge are called:

a) Dharma

b) Pramāṇa

c) Karma

d) Mokṣa

2Which of the following is accepted as a Pramāṇa by almost all Indian philosophical schools?

a) Arthāpatti

b) Anupalabdhi

c) Pratyakṣa

d) Śabda

3Which Pramāṇa refers to knowledge gained through inference?

a) Pratyakṣa

b) Anumāna

c) Upamāna

d) Śabda

4Buddhism accepts how many Pramāṇas?

a) One

b) Two

c) Three

d) Four

5 Which two Pramāṇas are accepted by Buddhism?

a) Perception and Testimony

b) Inference and Comparison

c) Perception and Inference

d) Testimony and Presumption

Unleash your potential

1 What does Buddhist epistemology entail?

2 What are the primary sources of knowing (pramāṇas) recognised in Buddhist epistemology?

CHAPTER II
Nyaya Epistemology

Introduction

The Vaishesika system is chronologically subsequent to Sähkhya and predates Nyäya. It is likely antecedent to, and absolutely not subsequent to, Buddhism and Jainism. The term originates from 'Vishesa', meaning particularity, a distinguishing trait, or distinctiveness. The Vaishesika philosophy is a pluralistic realism that underscores the notion that diversity constitutes the essence of the universe. This theory extensively examines the concept of Vishesa, or particularity, which is considered the essence of entities. The originator of this technique is Kanada, sometimes referred to as Kanabhuk, Ulüka, and Kâshyapa. This technique is often referred to as 'Kanada' or 'Aulüka Darshana'. He was named Kanada due to his lifestyle as an ascetic, subsisting on grains collected from the fields. 'Kana', while signifying 'grain', also denotes a particle or a particular, whereas the term 'Kanada' implies an individual who adheres to the philosophy of particularity—vishesa. Prashastapâda authored the renowned Padärthadharmasahgraha, referred to as a Bhäsya or Commentary on Kanada's Vaishesikasütra, yet it constitutes a highly valued standalone work. Udayana and Shridhara have commented on it. The Vaishesika was subsequently integrated with the Nyäya, which embraced the ontology of the former and elaborated upon it through its epistemological framework. Consequently, Shivädítya, Laugäksi Bhaskara, Vishvanätha, and Annambhatta address the two systems concurrently, analysing their interconnections and the implications of their combined philosophical perspectives on knowledge and reality.

The philosopher Gotama is the originator of the Nyaya School. He is also referred to as Gautama and Aksapada. Nyäya denotes arguments and implies that the system is primarily intellectual, analytical, logical, and epistemological. It is also referred to as Tarkashästra, the science of reasoning; Pramänashästra, the science of logic and epistemology; Hetuvidyä, the science of causes; Vädavidyä, the science of argument; and Ânvïksikï, the science of critical examination.

Nyāya is a framework of atomistic pluralism and logical realism. It is associated with the Vaishesika system, which is considered 'Samänatantra' or a comparable philosophy. Vaishesika

formulates metaphysics and ontology; Nyäya constructs logic and epistemology. Both concur in perceiving earthly existence as replete with misery, constituting a bondage of the soul, and in considering liberation, defined as the complete cessation of suffering, as the ultimate objective of life. Both concur that bondage arises from ignorance of reality, whereas release stems from accurate understanding of reality. Vaishesika addresses the elucidation of reality, while Nyäya focuses on the articulation of accurate understanding of reality. Nyāya predominantly endorses Vaishesika metaphysics. However, several significant distinctions between them warrant attention. Initially, the Vaishesika identifies seven categories and categorises all realities inside them, whereas the Nyäya delineates sixteen categories and encompasses all seven Vaishesika categories under one, termed 'Prameya' or 'the Knowable', which is the second category among the sixteen. The initial category is Pramäna, denoting the valid means of knowing. This distinctly highlights the primarily logical and epistemological nature of the Nyäya philosophy. Secondly, while the Vaishesika acknowledges solely two Pramänas—perception and inference—dismissing comparison and verbal authority as forms of inference, the Nyäya distinguishes all four as separate entities: perception, inference, comparison, and verbal authority. Knowledge (jnana) is defined as apprehension (upalabdhi) or consciousness (anubhava). Nyāya posits that knowledge elucidates both the subject and the object, which are fundamentally separate from it, thereby allowing individuals to understand the relationship between their perceptions and the external world. All knowledge is a revelation or manifestation of objects (arthaprakäsho buddhih). Similar to how a lamp illuminates tangible items positioned before it, knowledge unveils all entities that present themselves to it. Knowledge can be either valid or invalid. Valid knowledge (pramä) is defined as the accurate comprehension of an object (yathärthänu- bhavah). It is the representation of a thing in its true form. Nyäya upholds the connection theory. For knowledge to be valid, it must align with reality. Valid knowledge is generated by the four legitimate means of knowledge: perception, inference, comparison, and witness. Invalid knowledge encompasses recollection (smrti), doubt (samshaya), error (viparyaya), and hypothetical reasoning (tarka). Memory is invalid, as it constitutes representational cognition rather than presentative cognition. The object of remembrance is not immediately accessible to the soul but is only recalled indirectly. Doubt constitutes doubt in thought. Error arises from a misunderstanding, as it fails to align with the actual item. Hypothetical reasoning does not constitute genuine knowledge. It posits the argument, "If there is no fire, then there cannot be smoke." Recognising a rope as such demonstrates accurate understanding. If you are unsure if it is a rope or a snake, you may

harbour doubts. If you remember the rope you have observed, you possess memory. To confuse the rope with a snake constitutes an error.

Knowledge is generated within the soul through its interaction with the non-soul. This characteristic of the soul is incidental and is produced by the object interacting with it. If the conditions for generation are robust, knowledge is legitimate; if they are flawed, knowledge is invalid. A man with normal vision perceives a conch as white, whereas a man afflicted with jaundice perceives it as yellow. The essence of truth lies in its correspondence with the object. Knowledge is legitimate if it aligns with its object; if it does not, it is invalid. Valid knowledge aligns with its object (yathārtha and avisamvādi) and facilitates effective action (pravṛttisāmarthya). Erroneous knowledge fails to align with its object, resulting in failure and disappointment (pravrttivisamväda). Fire must burn, cook, and illuminate. If it fails to ignite, it is not a fire. Knowledge is fundamentally a manifestation of objects. The inquiry of its validity or invalidity is a subsequent matter that is contingent upon its alignment with its aim. Truth and falsehood are external attributes of knowledge. They are seized by subsequent understanding. They emerge and are understood only after knowledge has already manifested. They are neither inherent nor self-apparent. The validity and invalidity of knowledge emerge (utpattau paratah prämänyam) subsequent to the acquisition of knowledge, and they are recognised (jnaptau paratah prämänyam) following the emergence of knowledge and their occurrence. Correspondence constitutes the substance, whereas successful action serves as the criterion for truth, indicating that a belief is true if it corresponds to reality and leads to effective outcomes in practice. The Nyäya theory of knowing is thus both realistic in its understanding of nature and pragmatic in its criteria for truth. The four types of valid knowledge are perception, inference, comparison or analogy, and verbal witness.

Perception

Gotama characterises perception as 'accurate cognition arising from the interaction of the sense organs with objects, devoid of nomenclature, and distinctly delineated.' "This concept of perception omits divine and yogic perception, which is not produced by the interaction of the sense organs with external objects. Vishvanätha defines perception as 'direct or instantaneous cognition that is not obtained through the mediation of any other cognition. ' This concept encompasses both ordinary and extraordinary perception while excluding inference, comparison, and witness. Perception constitutes a form of knowledge and is an inherent characteristic of the self. Conventional perception assumes the presence of sense organs, objects, the mind, and the self, together with their interactions. The self interacts with the manas, the manas with the sensory organs, and the sensory organs with the objects. Contact

between the sense organs and things is contingent upon the manas first engaging with the sense organs, which in turn requires the self to interact with the manas. Therefore, interaction between sense objects inherently requires both contact between the manas and the senses, as well as contact between the self and the manas. The sensory organs originate from elements that exhibit unique attributes of smell, taste, colour, touch, and sound, which are essential for the process of perception and interaction with the environment. Perception constitutes a form of knowledge and is an inherent characteristic of the self. Conventional perception assumes the presence of sensory organs, objects, the mind, and the self, together with their interactions. The self interacts with the manas, the manas with the sensory organs, and the sensory organs with the objects. The interaction of the sense organs with objects is contingent upon the prior engagement of the manas with the sense organs, which in turn requires the self to first connect with the manas. Therefore, contact between sense objects inherently requires the interaction between the mind and the senses, as well as the interaction between the self and the mind. The sensory organs originate from elements that exhibit unique attributes for smell, taste, colour, touch, and sound. The manas serves as the intermediary between the self and the sensory organs, facilitating the processing and interpretation of sensory information to form perceptions and experiences. The external object, perceived through the senses and the manas, creates an impact on the self. Consequently, the theory is pragmatic.

The Naiyāyika asserts two phases in perception. The first is termed 'indeterminate' or 'nirvikalpa', whereas the second is referred to as 'determinate' or 'savikalpa'. They represent not two distinct forms of vision, but rather the initial and subsequent phases within the same intricate process of perception. Gotama acknowledges these two levels in his aforementioned description of perception. Perception is 'unassociated with a name' (avyapadeshya), signifying 'indeterminate', and it is 'well-defined' (vyavasayatmaka), indicating 'determinate'. All perception is definitive, but it is invariably preceded by a prior phase of indeterminacy, which allows for the initial gathering of sensory information before it is processed into a clear understanding. Nvâva acknowledges the essential principle about knowledge, attributed as Kant's unique contribution to Western philosophy, which asserts that knowledge encompasses both sense and idea. “Percepts devoid of conceptions are blind, whereas concepts devoid of percepts are vacuous.” Perception is a multifaceted process of experience that encompasses both sensation and cognition. All perception is determined, as it constitutes perceptual knowledge or judgement. Sensation constitutes the substance, while idea represents the structure of knowledge. Bare sensation or simple apprehension constitutes nirvikalpa perception, while perceptual judgement or relational apprehension represents savikalpa

perception. Nyāya circumvents the mistake of the psychical staircase hypothesis, which posits that one first has sensory experience, followed by conceptualisation and subsequently judgement. Perception is a multifaceted process of presentation and representation, wherein direct consciousness cannot be distinctly separated from relational judgement. Indeterminate perception constitutes the substance from which determinate perception is formed; yet, they can only be differentiated conceptually and cannot be separated in actuality. Nirvikalpa perception is the initial apprehension, pure consciousness, and direct sensory experience that is undifferentiated, non-relational, and devoid of assimilation, discrimination, analysis, and synthesis. The awareness of "that" is not yet defined by the awareness of "what". However, since the 'that' cannot be comprehended independently from the 'what', and the 'substance' cannot be understood apart from its 'qualities', we promptly arrive at savikalpa perception. Here, the mere awareness of the 'that' and the 'what', along with their 'inherence' as an undifferentiated, unrelated, mute, and inarticulate entity, is transmuted into differentiated, relational, conceptual, and articulate knowledge that encompasses assimilation, discrimination, analysis, and synthesis. For instance, upon entering a dark cinema hall from bright daylight to attend a matinée, our initial view of the seats and audience is obscured, presenting only a vague awareness of the objects, which progressively become discernible; this indistinct perception of the objects in the hall constitutes indeterminate perception, whereas the subsequent clear recognition of them represents determinate perception. The simple recognition of an object as a 'that' is indeterminate perception, but the distinct perception of it along with its characteristics represents determinate perception. At dusk, we observe a linear object on the road and ascertain upon closer inspection that it is a rope. A distant white object approaches, revealing itself to be a white cow. The initial phase is ambiguous, while the subsequent phase is characterised by determined perception. We are pressed for time and wish to complete our bath before departing. It is uncertain whether the water was cold and the bath invigorating; however, we did perceive the water's coolness and the bath's refreshing nature. We perceive water and its coolness, although we do not associate the two sensations. Indeterminate perception reveals the thing in its unadorned state, devoid of any characteristics. In determinate perception, we associate the substance with its properties. The sensation of ambiguous perception is psychological, while its understanding is logical. In its purest form, as just awareness, we perceive an indeterminate sensation; we experience it, yet the instant we attempt to comprehend it as 'bare awareness', it transitions into conception and becomes determinate. Therefore, all our perceptions, as forms of cognition, are determinate and constitute perceptual judgements. Indeterminate perception may only be conceptually distinguished from determinate perception, not in actuality, which

means that in practice, our experiences of indeterminate perception are often influenced by prior knowledge and contextual factors that shape our interpretations. Therefore, while we perceive ambiguous perception as a psychological state of sensory experience, our understanding of it, even as indeterminate perception, arises from logical deduction. We experience it directly, although it is just awareness, not cognition. Pure apprehension, being an infra-relational phenomenon, cannot be comprehended without the context of cognitive processes that follow, which transform it into meaningful experiences. Cognition is inferred thereafter when conceptualisation has converted a mere sensation into a perceptual judgement.

Vātsyāyana asserts that when an item is recognised by its name, it results in determinate perception; conversely, when observed without its name, it leads to indeterminate perception. Jayanta Bhatta asserts that indeterminate perception recognises material, characteristics, and acts, as well as universals, as independent and indistinct entities without any association with a name, whereas determinate perception comprehends all these elements in conjunction with a name. Gangesha Upädhyäya characterises indeterminate perception as the non-relational recognition of an item free from any association with a name, genus, or differentia. Annam Bhatta defines indeterminate perception as the direct perception of an item and its attributes, without understanding the relationship between the two. The 'that' and the 'what' are seen separately, and it is not understood that the 'what' characterises the 'that' or that these qualities inhere in the substance. 'Indeterminate perception' refers to 'mere acquaintance', which William James describes as 'raw, unverbalised experience', whereas 'determinate perception' denotes relational apprehension. Perception can be classified as either ordinary (laukika) or extraordinary (alau-kika). Laukika perception occurs when the sense organs interact with objects in a conventional manner. If the interaction between the sense organs and things occurs in an atypical manner, meaning the items are not normally accessible to the senses but are transmitted through an extraordinary channel, we experience Alaukika perception. Perception is categorised into two types: internal (mänasa) and outward (bähya). The mind (manas), as the internal organ, engages with psychical states and processes like cognition, emotion, volition, desire, pain, pleasure, and aversion. External perception occurs when the five sensory organs interact with external objects. There are five types: visual, auditory, tactile, gustatory, and olfactory, which are elicited by the sensory organs of sight, sound, touch, taste, and smell, respectively, upon interaction with external objects. The external sense organs consist of the material elements of earth, water, fire, air, and ether, and so each perceives the specific quality

of its corresponding element. Consequently, the olfactory sense organ consists of terrestrial atoms and detects odours, which represent the distinctive quality of earth, and so on.

Extraordinary perception is categorised into three types: sämänyalaksana, jnänalaksana, and yogaja. Sämänyalaksana perception refers to the recognition of universals. Nyaya posits that universals constitute a separate category of realities. They reside in the specifics associated with various categories due to the distinct universals inherent in them. An individual is classified as belonging to a specific category due to the inherent qualities of that category residing inside it. Consequently, a cow is classified as a cow due to the inherent quality of universal cowness within it. Typically, we recognise only the specifics and not the generalities. We recognise specific cows, but we do not sense a 'universal cow'. Thus, the Nyäya asserts that universals are sensed in an exceptional manner. When we observe a specific cow, we initially recognise the inherent 'universal cowness' inside it. The second type of unusual perception is referred to as jnänalaksana perception. It is the intricate perception derived from association. Occasionally, disparate sensations get linked, resulting in a unified perception, which can lead to a more complex understanding of an experience or concept. An item is not directly viewed by a sense organ; rather, it is recalled from memory through prior cognition and is apprehended via representation. For instance, I observe a blossoming rose from afar and state, ‘I perceive a fragrant rose. 'However, how can scent exist? It can just be perceived through smell. Fragrance is detectable via the olfactory sense, whereas the visual sense is limited to perceiving colour. The visual perception of the rose evokes the recollection of its aroma through associative recall, previously experienced through olfactory senses. The experience of the aromatic rose through sight is referred to as 'jnänalaksana perception', or the recollection of the item based on prior cognition (jnäna). Additional instances include 'the piece of sandalwood appears aromatic’, ‘ice appears frigid’, ‘stone appears solid’, ‘tea appears steaming’, etc. The Nyäya theory of illusion, termed 'Anyathäkhyäti', is founded on this type of perception. When we confuse a rope with a snake, the concept of the snake, previously seen, is recalled by this remarkable jnänalaksana perception and is conflated with the object (i.e., rope) that is immediately perceived by the sense organ. When a shell is erroneously identified as silver, the memory of silver previously encountered at a shop (äpanastha) or elsewhere is reactivated by jnanalaksana perception and conflated with the item (i.e., shell) that is directly received by the sense organ. The past impression signifies the object in our consciousness. The error arises from an incorrect synthesis of the presented and represented objects. The depicted object is mistaken for the displayed one. The term 'anyathä' signifies 'otherwise' and 'elsewhere', and both meanings are

manifested in a mistaken perception. The exhibited object is interpreted differently, while the represented thing exists in a different location. The shell, the silver, the rope, and the snake all exist independently; their combination is merely an illusion. The shell and the rope are explicitly identified as 'this' (when we state: 'this is silver' or 'this is a snake'), whereas the silver and the snake reside in a different realm and are recalled through jnänalaksana perception. The third type of exceptional perception is termed 'yogaja perception'. The Yogins gain an intuitive and immediate perception of all objects—past, present, and future—through the power of meditation. It resembles the Kevalajnana of the Jains, the Bodhi of the Buddhists, the Kaivalya of the Särtkhya-Yoga, and the Aparoksanubhuti of the Vedantins. It is instinctive, transcendent, and beyond relational dynamics.

Inference

The second type of knowledge is anumä, which is inferential or relational, and its method is referred to as anumäna, or inference. It is characterised as a cognition that presupposes another cognition. It is mediated and indirect, arising through a 'mark', the 'middle term' (linga or hetu), which is inevitably associated with the 'main term' (sädhya). It is knowledge (mäna) that emerges after (anu) other knowledge. Invariable concomitance (vyâpti) is the essence of inference. The inclusion of the middle term within the minor term is referred to as 'paksadharmatä'. The constant correlation of the middle term with the major phrase is referred to as 'vyâpti'. The understanding of paksadharmatä, as defined by vyâpti, is referred to as parämarsha. Inference is described as knowledge derived from parämarsha, namely the awareness of the major in the minor through the intermediary that exists inside the minor (paksa-dharmatä) and is consistently linked to the major (vyäpti). Similar to the Aristotelian syllogism, Indian inference comprises three terms. The major, minor, and middle are referred to as 'sädhya', 'paksa', and 'linga' or 'hetu', respectively. Smoke is consistently linked to fire (vyäpti), and upon observing smoke on a hill, we deduce the presence of fire in that location. Hill constitutes the minor term; fire represents the major phrase; smoke serves as the middle term. Based on the observation of smoke on the hill and the principle that smoke indicates the existence of fire, we deduce that fire exists on the hill. This constitutes inference. Indian logic does not distinguish between reduction and induction. Inference is a multifaceted

process that encompasses both elements. Indian logic also refutes the verbalist perspective on logic. It examines cognition in its entirety rather than merely its manifestations. The formal and material logic are integrated here. The verbal form does not constitute an essential component of the inference. This is evident from the classification of inference into svärtha (for oneself) and parärtha (for others). In the former, a formal explanation of the various components of inference is unnecessary. It constitutes a psychological process. The parärtha, which is a syllogism, must be articulated in language solely to persuade others. The Nyāya syllogism comprises five components. The initial term is referred to as 'Pratijnä' or 'proposal'. This is the logical proposition that requires proof. The second is Hetu, or 'reason', which articulates the rationale for the establishment of the proposition. The third is termed Udäharana, which provides universal concomitance along with an illustration. The fourth is Upanaya, referring to the application of universal concomitance to the current issue. The fifth is Nigamana, or the conclusion derived from the prior assertions. The five assertions of the Indian syllogism are referred to as 'members' or 'avayavas'. A standard Nyāya syllogism is presented as follows:

This hill possesses fire (pratijnä).

(2) Due to the presence of smoke (hetu).

(3) Wherever there is smoke, there is fire; for instance, an oven (udäharana).

This hill emits smoke that is consistently linked to fire (upanaya).

Consequently, this hill possesses fire (nigamana).

When compared to the Aristotelian syllogism, which consists of three propositions, the Nyāya syllogism aligns with the Barbara (AAA) mood of the first figure, recognised as the most robust mood of the strongest figure. Although the Nyäya syllogism comprises five assertions and the Aristotelian syllogism consists of three, both use only three terms: sädhya (major), paksa (minor), and hetu (middle). Among the five propositions, two seem superfluous, allowing us to omit either the initial two or the last two, as they are fundamentally identical. The first aligns with the fifth, and the second corresponds with the fourth. Excluding the last two, the initial three propositions align with the conclusion, the minor premise, and the major premise, respectively. If we exclude the first two, the final three assertions align with the major premise, the minor premise, and the conclusion of the Aristotelian syllogism. Consequently, if we

exclude the initial two components of the Nyāya syllogism, which are encompassed inside the final two, we observe that it bears resemblance to the Aristotelian syllogism. First Figure:

(1) All things which have smoke have fire (major premise).

(2) This hill has smoke (minor premise).

(3) Therefore this hill has fire (Conclusion).

And the typical Aristotelian syllogism may be stated in the Nyäya form thus:

(1) Socrates is mortal (pratijnä).

(2) Because he is a man (hetu).

(3) Whoever is a man is a mortal, e.g., Pythagoras (udäharana).

(4) Socrates is a man who is invariably a mortal (upanaya).

(5) Therefore Socrates is mortal (nigamana).

However, there exist distinct discrepancies between the Nyäya and Aristotelian syllogisms, beyond the mere notional variation in the number of propositions in each. The Aristotelian syllogism is only deductive and formal, whereas the Nyäya syllogism is both deductive-inductive and formal-material. The Nyäya appropriately perceives deduction and induction as intrinsically connected, representing two facets of the same process—an understanding now acknowledged in Western logic. Inference, as per Nyäya, is derived not from the universal to the particular or vice versa, but rather from the specific to the particular via the universal. This example exemplifies a distinctive characteristic of the Nyäya syllogism, demonstrating that the universal major premise arises from genuine induction grounded in the rule of causation and that induction and deduction are inherently interconnected. In the Aristotelian syllogism, the main and minor words are distinct in the premises; however, they are linked by the middle term. Conversely, in the Nyäya syllogism, all three elements are integrated inside the Upanaya. While the Aristotelian syllogism is primarily verbal, the Nyäya acknowledges that verbal expression is not the substance of inference and is merely necessary for persuading others. There exist distinct basic disparities between the two perspectives. We more reasonably attribute the similarities between the two to a parallel evolution of thought. Indian logic has developed organically.

There are five characteristics of the middle term:

(1) It must be present in the minor term (paksadharmatâ); e.g., smoke must be present in the hill.
(2) It must be present in all positive instances in which the major term is present; e.g., smoke must be present in the kitchen where fire exists (sapaksasattva).
(3) It must be absent in all negative instances in which the major term is absent; e.g., smoke must be absent in the lake in which fire does not exist (vipaksäsattva).
(4) It must be non-incompatible with the minor term; e.g., it must not prove the coolness of fire (abädhita).
(5) It must be qualified by the absence of counteracting reasons which lead to a contradictory conclusion; e.g., ‘the fact of being caused’ should not be used to prove the ‘eternality of sound' (aviruddha).

Inference is typically classified into two types: Svärtha and Parärtha, which we have previously examined. Gotama delineates three categories of inference: pûrvavat, shesavat, and sämänyatodrsta. The initial two are predicated on causation, whereas the final one is founded on mere coexistence. A cause is the constant and unconditional precursor of an effect, while an effect is the constant and unconditional result of a cause. Pûrvavat inference occurs when we deduce an unobserved result from a visible cause, such as inferring impending rain from the presence of dark clouds in the sky. When we deduce an unobserved cause from a perceived consequence, we engage in shesavat inference; for instance, inferring past rainfall from the rapidly flowing, muddy, and inundated water of a river. Inference derived from regularity of co-existence rather than causality is termed 'sämänyatodrsta'; for instance, inferring the presence of cloven hooves in an animal based on its horns. One interpretation posits that a pûrvavat inference relies on prior experience of universal concomitance between two entities, a shesavat inference is characterised as 'parishesa' or inference by elimination, and a sämänyatodrsta is defined as inference by analogy.

Another classification of inference includes kevalänvayi, kevala vyatireki, and anvaya vyatireki inferences. It is based on the essence of vyäpti and the many methodologies for its establishment. The methods of induction that demonstrate universal causal relationships may include anvaya, vyatireka, or a combination of both. The first pertains to Mill's Method of Agreement, the second to his Method of Difference, and the third to his Joint Method of Agreement and Difference, often known as the Method of Double Agreement. Kevalänvayi inference occurs when the middle term is consistently positively correlated with the major

word. The concepts concur solely in presence, with no negative instance of their concurrence in absence, e.g.,

All knowable objects are nameable;

The pot is a knowable object;

The pot is nameable.

We have kevalavyatireki inference when the middle term is the differentium of the minor term and is always negatively related to the major term. The terms agree only in absence, there being no positive instance of their agreement in presence, e.g.,

What is not different from other elements has no smell;

The earth has smell;

The earth is different-from-other-elements.

We have anvayavyatireki inference when the middle term is both positively and negatively related to the major term. The vyäpti between the middle and the major is in respect of both presence and absence. There is double agreement between the terms—they agree in presence in the positive instances, and they also agree in absence in the negative instances; e.g.,

All things which have smoke have fire;

This hill has smoke;

.'. This hill has fire; and

No non-fiery things have smoke;

This hill has smoke;. This hill is not non-fiery; i.e., This hill has fire.

In Indian logic a fallacy is called 'hetvabhasa'. It means that the middle term appears to be a reason but is not a valid one. All fallacies are material fallacies. We have mentioned the five characteristics of a valid middle term. When we violate these characteristics, we encounter fallacies. Five kinds of fallacies are recognised:

(i)Asiddha or Sädhyasama: This is the fallacy of the unproved middle. The middle term must be present in the minor term (paksadharmatä). If it is not, it is unproved. It is of three kinds—

(a) äshrayä siddha: The minor term is the locus of the middle term. If the minor term is unreal, the middle term cannot be present in it; e.g., 'the sky-lotus is fragrant, because it is a lotus, like the lotus of a lake'.

(b) svarüpä siddha: Here the minor term is not unreal. But the middle term cannot by its very nature be present in the minor term; e.g., 'sound is a quality because it is visible'. Here visibility

cannot belong to sound which is audible

(c) vyäpyatvä siddha: Here vyâpti is conditional (sopädhika). We cannot say, e.g., 'Wherever there is fire, there is smoke.' Fire smokes only when it is associated with wet fuel. A red-hot iron ball or clear fire does not smoke. Hence, 'association with wet fuel' is a condition necessary to the aforesaid vyâpti. Being conditioned, the middle term becomes fallacious if we say, 'The hill has smoke because it has fire.'

(2) Savyabhichära or Anaikäntika: This is the fallacy of the irregular middle. It is of three kinds.

: (a) Sädharana: Here the middle term is too wide. It is present in both the sapaksa (positive) and the vipaksa (negative) instances and violates the rule that the middle should not be present in the negative instances (vipak-säsattva); e.g., 'the hill has fire because it is knowable'. Here 'knowable' is present in fiery as well as nonfiery objects.

(b) Asädhärana: Here the middle term is too narrow. It is present only in the paksa and neither in the sapaksa nor in the vipaksa. It violates the rule that the middle term should be present in the sapaksa (sapaksasattva); e.g., 'sound is eternal because it is audible. Here, audibility belongs to sound alone and is present nowhere else.

(c) Anupasarhhäri: Here the middle term is non-exclusive. The minor term is all-inclusive and leaves nothing by way of sapaksa or vipaksa; e.g., 'All things are noneternal because they are knowable.'

(3) Satpratipakça: Here the middle term is contradicted by another middle term. The reason is counterbalanced by another reason. And both are of equal force; e.g., 'sound is eternal because it is audible' and 'sound is non-eternal because it is produced'. Here 'audible' is counterbalanced by 'produced', and both are of equal force.

(4) Bâdhita: It is the non-inferentially contradicted middle. Here the middle term is contradicted by some other pramäna and not by inference. It cannot prove the major term, which is disproved by another stronger source of valid knowledge; e.g., 'fire is cold because it is a substance. Here the middle term 'substance' becomes contradicted because its major term 'coldness' is directly contradicted by perception.

(5) Viruddha: It is the contradictory middle. The absence of the major term pervades the middle term, not its presence. Instead of proving the existence of the major term in the minor term, it proves its non-existence therein; e.g., 'sound is eternal because it is produced.' Here 'produced', instead of proving the eternality of sound, proves its non-eternality. Here the middle term itself disproves the original proposition and proves its contradictory, while in the savyabhichära the middle term only fails to prove the conclusion, and in the satpratipaksa the middle term is inferentially contradicted by another middle term, both of which are of equal force, and in the bâdhita the middle term is non-inferentially contradicted and the major is disproved by a stronger pramäna other than inference.

Comparison

The third type of legitimate cognition is called Upamiti, and the corresponding means of obtaining it is referred to as Upamâna. This type of cognition involves information that is obtained through comparison and is approximately aligned with analogy. It is described as understanding the relationship between a word and its denotation. It is generated by the understanding of likeness or similarity. A guy who is unfamiliar with a gavaya, or wild cow, learns that a wild cow resembles a cow and later encounters one in a jungle, identifying it as such; his understanding is attributed to upamâna. He has encountered the term 'gavaya' and has been informed that it resembles a cow; now he perceives the thing signified by 'gavaya' and acknowledges its resemblance. Thus, upamâna refers to the understanding of the relationship between a name and the object it signifies. The recognition of similarity produces this understanding, as an individual identifies a wild cow as a 'gavaya' upon perceiving its resemblance to a cow and recalling the description that defines a 'gavaya' as an animal akin to a cow. Buddhists construe Upamäna as perception and testimony. The Sänkhya and Vaishesika schools limit it to inference. The Jainas simplify it to recognition or pratyabhijñā. The Mimâmsakas acknowledge it as an independent source of knowledge; nonetheless, their interpretation diverges from that of Nyäya, which will be examined in the chapter on Mimämsä.

Verbal Testimony

The fourth category of genuine knowledge is Shabda, Agama, or authoritative verbal testimony. Its significance is often referred to as Shabda. It is characterised as the assertion of a reliable individual (aptavakya) and entails comprehending its significance. A phrase is characterised as a compilation of words, whereas a word is described as an entity capable of conveying meaning. The ability of a word to express its meaning originates, as per old Nyāya, from God, and according to later Nyāya, from long-standing convention. Testimony is inherently subjective. It is founded on the statements of a reliable individual, whether human or divine. Testimony is categorised into two types: Vaidika and laukika (secular). The Vaidika witness is flawless and infallible, as the Vedas are divinely articulated; in contrast, secular testimony, being articulated by flawed humans, lacks infallibility. Only the statements of reliable individuals who consistently convey the truth hold validity; those of others do not. A word is a powerful symbol representing an object, while a sentence is a compilation of words. A phrase must adhere to specific requirements to be comprehensible. There are four conditions: äkänksä, yogyatä, sannidhi, and tätparya. The initial concept is mutual implication or anticipation. The words in a phrase are interconnected and depend on each other to convey a complete meaning. An arbitrary collection of disjointed words will not constitute a coherent sentence. It will be utter nonsense, for example, 'cow horse man elephant'. The second criterion is that the language must be appropriate to communicate the intended meaning and must not contradict it. "Watering the plants with fire is a paradoxical statement." The third criterion is the close proximity of the words to each other. The words must be articulated in rapid succession without prolonged pauses. If the words 'bring', 'a', and 'cow' are articulated at extended intervals, they would not constitute a coherent phrase. The fourth requirement pertains to the speaker's intention when the words are uncertain. The term 'saindhava' signifies both 'salt' and 'horse'. When a man consuming food requests another to bring 'saindhava', the latter should not present a horse.

The Nyäya recognises solely these four pramänas. Arthāpatti, or implication, is distilled to inference. For instance, when we state, 'Fat Devadatta does not consume food during the day,' it implies that he must be eating at night; otherwise, how could he maintain his weight? Mïmâmsâ confers the status of an autonomous pramāṇa to implication. However, Nyäya simplifies it to inference as follows:

All fat persons who do not eat during the day eat during the night.
Devadatta is a fat person who does not eat during day;
Devadatta is a fat person who eats during the night.

Abhâva, or non-existence, which also is regarded as a separate pramäna by Bhâtta Mïmâmsâ, is reduced here either to perception or to inference. Abhâva is the non-existence of a thing, and the same sense organ which perceives a thing also perceives its non-existence. If the thing is imperceptible and can only be inferred, then its non-existence too may be equally inferred.

Nyaya epistemology makes a substantial contribution to Indian philosophy by providing a systematic and logical framework for the attainment of genuine knowledge. The Nyaya school emphasises the importance of rational inquiry, critical thinking, and logical analysis in the pursuit of truth. It clarifies that genuine knowledge (prama) is obtained from reliable sources known as pramanas. Nyaya epistemology clarifies the sources of valid knowledge and promotes a rational and analytical method of reasoning. The emphasis on logic, evidence, and critical reasoning has significantly influenced Indian philosophical traditions and is vital for understanding the essence and attainment of knowledge.

In the fifth generation of computers, considerable effort has been dedicated to simulating intelligent activity in machines. Researchers are endeavouring to address this issue by employing grammar-based computing, which aims to impart Knowledge of Language (KOL) to machines for Natural Language Understanding (NLU) or Processing (NLP), alongside knowledge-based computing, which seeks to represent background/contextual or Real-World Knowledge (RWK) in machines for comprehending real-world scenarios and problem-solving. The researchers are contemplating both options. A growing number of individuals are recognising that RWK representation is essential for the simulation of consciousness in machines. No comprehensive framework has been produced that is applicable to all forms of

knowledge, identities of the location, object, and instrument of cognition. Numerous scholars in the domain of artificial intelligence are starting to interrogate the veracity of the following assertions:

* Knowledge can be represented
* Computers can encode and decode knowledge in everyday language
* Computers are capable of thinking and understanding
* Computers require knowledge in order to communicate with people or other machines

While Panini's grammar is indeed relevant to RWK, it mostly serves as a framework for KOL that can be utilised as a model for more languages. This is due to the unequivocal accounts of both the KOL and the RWK in India, especially in the Sanskrit language. One can locate an early account of RWK within the Naya system, which bears a resemblance to the views presented by AL philosophers. Consequently, it is imperative to examine the offerings of the Naya system. The objective is to compare and contrast the Indian epistemic tradition, particularly the Naya, with those linked with artificial intelligence.

There are two major fields of research in artificial intelligence: knowledge representation (KR) and information retrieval (IR). Knowledge can be characterised as objects and attributes, or even as an algorithm that contains objects and procedures, when approached from the perspective of artificial intelligence (AI). An interaction between "data" and "processes" or "abilities" and "memory storage" is stated to be the cause of the so-called "intelligent behaviour" that is observed. According to Jean-Raymond Abrial (1974), a database is a representation of the physical world as it is always changing. During a particular moment, the state of this model is a representation of the knowledge that it has obtained from the outside world. But knowledge is more than static encoding of facts in interacting with the world. The idea that knowledge of anything is the capacity to construct a mental model that accurately depicts the thing in question, as well as the actions that may be carried out by it and on it, is one of the fundamental premises of artificial intelligence (AI). Following this, individuals (or robots) are able to make predictions about what is likely to occur in the real world by testing actions on the model (Sowa in 1984). An earlier challenge faced by those working on artificial intelligence was determining whether to represent knowledge by using a declarative approach or a procedural method. For the purpose of knowledge representation, it has been finally

decided that the most appropriate strategy would be one that combines a declarative approach with a procedural approach. Predicate calculus, semantic nets, frames, conceptual graphs, scripts, semantic primitives, and neural nets are some of the other well-known techniques for KR. These techniques follow a wide variety of strategies for KR. In spite of this, reducing the unknown phenomenon known as knowledge to a single or two definitions would not be an aim in and of itself. According to Hopcroft and Lipman (1979), not only would it be necessary to comprehend the entirety of the human cognitive process, but it would also be necessary to develop efficient methods for representing information.

pratyaksa (perception)

* anumana (inference)

* upamana (analogy)

* sabda (verbal testimony) and latter is sub-classified as

* samsaya (doubt)

* viparyaya (error)

* tarka (hypothetical argument)

* smrti (memory)

5.1. **Comparison with AI**

The following sub-sections define each of the above sub-classifications of invalid and valid knowledge and give their AI parallels--

5.1.1. apramana--the invalid knowledge

It is defined as false knowledge, doubtful or illusionary. The four types of invalid knowledge are

defined and compared with AI as follows:

5.1.1.1. smrti (memory)

* is impressions left by previous knowledge

* it is non valid because the object of knowledge is absent at the time of remembrance

* AI explanation

* memory as gaps in knowledge or probabilistic knowledge which must be corrected before processing starts and conclusion is arrived at

* such gaps or incomplete knowledge may be replaced by new knowledge [right arrow] nonmonotonic knowledge (McDermott & Doyle 1980)

* static memory [right arrow] knowledge of object (WHAT)

* short term : memory in current focus

* long term: memory for future use

* procedural memory [right arrow] knowledge of behavior (HOW)

* short term

* long term

5.1.1.2. samsaya (d * conflicting judgment on the precise character of an object of cognition (Bijalwan 1987)

* neither true or false, only wavering judgment

* AI explanation

* ambiguity resolution and bug handling come close to samsya

5.1.1.3. tarka (hypothetical judgment)

* AI explanation

* Heuristics

* when reality is not known, truth is ascertained trying alternatives

* atmasraya (petito principi)

* if A [right arrow] A then A must be different from A

* anyonyasraya (dependent tarka)

* A[right arrow]B[right arrow]A

* cakraka (circular logic)

* recursion in AI

* A[right arrow]B[right arrow]C[right arrow]A

* anavastha (regressus ad infinitum or transitivity)

* A[right arrow]B[right arrow]C[right arrow]D ...

* tadanya-budhiat-prasanga (reduction ad absurdum)

* proves a proposition by negating the contradiction of the proposition

* A[right arrow]-(-A) or + [right arrow] -(-)oubt)

5.1.1.4. viparyaya (error)

* manifestation of a real object in another * positive misconception, incomplete knowledge, gaps in knowledge * AI explanation * can be compared to bugs and inconsistencies

5.1.2. pramana--the valid knowledge

The following section discusses the means to valid knowledge and compares them with AI

5.1.2.1. **pratyaksa (perception)**

* Five senses--sight, smell, taste, hearing, touch (mind [right arrow] sixth sense for cognition of
feelings like pleasure, pain etc)
* six ways of perception for six kinds of objects
* substance, quality, universal quality, sound, sound-ness, nonexistence
* AI [right arrow] icon, schema and percept
* icon [right arrow] temporary record of a sensory input that the brain keeps
* schema [right arrow] blue print of the mental model
* percept [right arrow] assembling units which are to be assembled according to the schema to
create a mental model
* Sowa (1984) calls the search mechanism "associative comparator" which must have the following
characteristics:
* associative retrieval
* brain is like a computer, retrieves data by an address in the storage by matching the best pattern
* top-down match
* perception searches for percepts that match the overall pattern of an icon
* stimulus constancy
* stimuli from same external object are recognized as equivalent despite varying size, brightness etc
* distributed storage
* images are not stored at a specific point, but distributed

5**.1.2.2. anumana (inference)**

* Gautama presupposes perception
* Jayanta gives five processes of inferring
* Perception of reason
* Remembrance of the universal concomitance
* Judgment that the subject of inference contains sense which, is concomitant with the object
* Knowledge of the consequence, and

* Judgment that the consequence is worthy of being accepted or rejected

* AI

* inference is very important and strong point of predicate calculus

* axioms [right arrow] theorems

* new facts can be deduced from axioms using rules of inference similar to ones given by Gautama

Gautama gives 3 ways to infer:

* psrvavat (deduction)

* cause [right arrow] effect.

* cloud [right arrow] rain

* "Gauri eats grass" from "Gauri is a cow" and "all cows eat grass" --

(inst gauri cow)

(forall X (if (inst X cow) (food X grass)))

(food gauri grass)

* sesavat (abduction)

* process [right arrow] explanations

* effect [right arrow] cause

* river is swollen [right arrow] there could have been rain

* Abduction has the following paradigms

From: b

(if a b)

Infer: a

* unlike deduction, abduction is not a legal interference. It can lead to false conclusions--

from: (feels nervous Ira)

(forall (X) (if (is sick X)

(feels nervous X)))

infer: (is sick Ira)

* samanyatod (induction)

* infers consequent from the antecedent, which is neither cause nor effect

* while induction can take several forms, the most common is

from: (P a), (P b),...

infer: (forall (X) (P X))

* although this is not a sound inference, induction like abduction, is very useful in everyday life where
it is more commonly known as learning
* for example, if we see a lot of leaves of green color, we might infer that all leaves are of green color
from: (if(inst leaf-1 leaf) (colour leaf_1 green))
(if(inst leaf-2 leaf) (colour leaf_2 green))
Infer: (forall (X) (if(inst X leaf) (colour X green)))
* Vacaspati divides inference into vita and avita (Bijalwan 1987). The former is based on universal
agreement in presence. As an example, we can infer
from: whatever has smoke has fire the hill has smoke
infer: the hill has fire
* avita is based on universal agreement in absence
from: whatever is non-different from other objects has no
smell. Earth has smell
infer: Earth is different from others.
* Prasastapada has svarthanumana and pararthanumana as two types of inference.
* Svarthanumana
* the premises are known from our own experience
*pararthanumana
* They are discovered by one man and imparted into another through language.
* Therefore, there is greater chance of error in pararthanumana which is based on words.svarthanumana is a mental process which is divided into data and samanyatodata by Prasastapada(Bijalwan 1987).
* Udyotakara gives kevalanvayin (K), kevalavyatirekin (KV), and amayavyatirekin (AV) as three types of inference. If a middle term is only positively related to the major term it is called K. Here. Thereason exists in the subject and similar instances, and is devoid of dissimilar instances. For example--we can infer "pot is nameable" from "pot is knowable" and "all knowable things are nameable"--
(property X nameable)))
(if (inst pot knowable)

(property pot nameable))

(property pot nameable)

In the case of KV, the middle term is negatively related to the major term. Reason exists in the subject, but not in dissimilar instances. For example--

from: (no (not P) is (M))

S is M

infer: S is P

When the middle is positively as well as negatively related to the major term, and reason exists in the entire subject, and in all similar instances but does not exist in dissimilar instances, the case is that of AV, as "sound is non-eternal because it is produced like a jar". The inference of Fire from smoke is also of this kind. Jayanta does not however, accept this division of inference in to K, KVand AV by Udyotakara (Bijalwan 1987).

5.1.2.3. upamana (analogy)

* upamana is the means by which we gain the knowledge of a previously unknown object on thebasis of its similarity to another object previously well known

* Mimansists and the Vedantists also accept analogy as an independent source of knowledge

* AI

* analogy is too slippery to be seen as just another road to induction

* often, analogy matches are so slipshod that they merely suggest solutions to the problem ofknowledge acquisition in general, and for inductive learning in particular

* analogy is useful because it serves as initial framework for developing the new concepts

5.1.2.4. sabda (verbal testimony)

* Leaving aside the Carvakas, Vaisesikas and the Buddhists, all the other systems of Indian philosophy accept sabda as a distinct source of knowledge, but they also differ with one another with respectto its nature forms and a number of other aspects.

* Jayanta (Nyayamanjari)

* contains a long discussion on this problem

* justices the acceptance of sabda as a distinct means of knowledge

* gives serious thought to the philosophy of language

* tries to explain the views of his predecessors and evaluates

* the arguments of prominent scholars

* his account of sabda is as much relevant to grammar, rhetoric, and linguistics as it is to logic

* convincingly proves that words do not exist before their production
* also refutes the sphota theory of grammarians and argues that the relationships between words
and meaning is conventional
* AI
* sabda can be compared to the concept of formal string and language.
* language is a set of strings made from some alphabet
if a fixed alphabet = E, then the language formed by
E is called [E.sup.*]
ifE= {a} then
[E.sup.*] = {e. a. aa. aaa. ...}

The Naya system and the AI system are comparable and contrastable. The Naya system exhibits numerous parallels with AI regarding its management of valid and invalid types of knowledge. The AI utilises various methodologies beyond logic and inference. These methodologies, such as semantic networks, conceptual graphs, frames, rule-based systems, and processes, provide recommendations for effectively representing information. This contrasts with the Naya and Western schools, which excessively emphasise rectifying the identity of the locus of knowledge acquisition. The latter may currently hold little significance for artificial intelligence. Nonetheless, there exists an urgent necessity to pursue further investigation into various Indian systems, including the neural network system and Buddhist logic, to formulate suitable models of human cognition applicable to artificial intelligence.
Nyāya's method of distinguishing valid from invalid information parallels the approach of artificial intelligence in differentiating correct predictions from inaccuracies. False inference, or mithyājñāna, can be likened to algorithmic bias or misclassification in artificial intelligence (AI). The Nyāya methodology rectifies faults by reasoning, akin to the training of artificial intelligence models, where iterative learning diminishes inaccuracies. To verify truth claims, artificial intelligence could employ logical validation akin to Nyāya. Contemporary artificial intelligence mistake-detection frameworks may draw inspiration from Nyaya's categorisation of perceptual and inferential faults utilised in error analysis. Hierarchical knowledge graphs in artificial intelligence exhibit similarities to the structured cognitive method of Nyāya.

Collaborative Reasoning between Humans and Artificial Intelligence: The integration of human logical reasoning (Nyāya) with machine inference (AI) offers the potential to enhance interpretability and accountability. The logical framework of Nyāya emphasises the cause-effect relationship, termed vyāpti, which asserts that a specific cause invariably produces a specific outcome. Smoke indicates the presence of fire. In the realm of artificial intelligence (AI), this parallels the notion of causal reasoning and explainable AI (XAI), which entails guaranteeing that an AI's output is not just statistically valid but also logically and causally justified. This concept is fundamental to Nyāya logic.

In comparing the Nyāya system with contemporary artificial intelligence (AI) systems, it is crucial to recognise that each system adopts a distinct approach to knowledge. The neural network system exhibits notable similarities with artificial intelligence (AI), especially in its systematic distinction between valid (pramā) and invalid (aprāmā) forms of knowledge. The two frameworks recognise that knowledge transcends mere information acquisition; it must possess specific attributes of trustworthiness, accuracy, and justification to be deemed knowledge. Conversely, although the Naya system relies predominantly on structured logical reasoning and accurately defined epistemic categories, modern AI systems employ a broader array of methodologies that extend beyond conventional logic and inference alone.The importance of Indian epistemological frameworks for artificial intelligence research remains undiminished. Conversely, there remains a substantial necessity to connect with Indian philosophical traditions, including the Naya system and Buddhist logic, in a more profound manner to cultivate models of cognition that are more intricate and nuanced. Buddhist theories of inference, momentariness, and dependent origination offer alternative frameworks for elucidating thinking processes. These theories do not assume the presence of a permanent self, aligning with the non-subjective nature of artificial intelligence systems. The Naya system's thorough analysis of error, uncertainty, and validation can yield significant insights for the advancement of artificial intelligence systems that are more robust, transparent, and epistemically responsible.

Knock your brain

1. Nyāya philosophy is primarily concerned with:

a) Liberation through devotion

b) Logical reasoning and valid knowledge

c) Ritual practices

d) Meditation techniques

2In Nyāya, valid knowledge is called:

a) Avidyā

b) Pramā

c) Samsāra

d) Mokṣa

3 How many Pramāṇas are accepted by the Nyāya school?

a) Two

b) Three

c) Four

d) Six

4 Which of the following is NOT a Nyāya Pramāṇa?

a) Perception (Pratyakṣa)

b) Inference (Anumāna)

c) Comparison (Upamāna)

d) Presumption (Arthāpatti)

5 Anumāna (inference) is based on:

a) Direct experience

b) Testimony

c) Universal relation (Vyāpti)

d) Guesswork

Unleash your potential

1.Elucidate the theory of knowledge and pramāṇas within the Nyaya framework?

2.Elucidate the Nyaya theory of inference, providing appropriate instances?

CHAPTER III

Western epistemology

Epistemology is defined as the branch of philosophy that addresses issues related to the origin, nature, validity, boundaries, and conditions of knowing. Epistemology probably started in Western philosophy when Greek sophists rigorously tested the ideas of their predecessors. The fundamental issues of epistemology were further developed within the Western philosophical tradition, particularly through the works of philosophers such as Descartes, Locke, and Kant, who explored the nature of knowledge, belief, and justification.

The Western viewpoint on knowledge has historically focused on the distinction between information that is acquired (a posteriori) and knowledge that is inherent (a priori), which means "from what is before" or "from what is after". In contrast to the latter's progress toward the empiricist method of inquiry, the former developed a rationalistic approach to examination, emphasising the role of reason and innate ideas in the acquisition of knowledge. Plato asserts that there exists a consciousness of absolute, universal conceptions or forms that exist independently of any person seeking to understand them. In his pursuit of knowledge, Aristotle emphasised logical and empirical methods. Furthermore, he recognises the perspective that knowledge constitutes comprehension of fundamental facts that are universally relevant.

What are the justifications for the existence of a discipline such as epistemology? Aristotle, who lived from 384 to 322 BCE, posits that the resolution lies in his claim that philosophy commences with a sensation of wonder or perplexity. The majority of individuals on Earth possess a desire to comprehend their surroundings, and many formulate various concepts to help interpret the reality they encounter. Conversely, because of the complexity of numerous aspects of the universe, most individuals are likely to abandon their efforts at some point and be content with the level of understanding they have achieved, often settling for simplified explanations or accepting uncertainty rather than pursuing more profound knowledge.

Philosophers, unlike other individuals, are captivated—some could even assert obsessed—by the notion of understanding the universe in the most expansive terms possible. Consequently, they strive to formulate theories that are both comprehensive and descriptively precise, explanatorily robust, and rationally justifiable in all aspects. In doing so, they advance the inquiry process beyond the typical approach, which is what is meant by the assertion that they formulate a philosophy on such subjects. Epistemologists, like most individuals, typically commence their theories with the assumption that they possess substantial information. As

individuals contemplate their expected knowledge, they increasingly recognise its diminishing reliability compared to their previous assumptions. They come to the awareness that numerous beliefs they once regarded as their most unwavering convictions are dubious or even false. These uncertainties stem from specific

Epistemology, alongside metaphysics, logic, and ethics, constitutes one of the four principal branches of philosophy. It is a philosophical branch focused on several epistemological hypotheses. It is a field that examines the nature, genesis, and boundaries of knowledge. It investigates the origin of knowledge and the criteria for its validity. The term 'epistemology' originates from the Greek words 'episteme' and 'logos', meaning 'knowledge' and 'reason', respectively. Epistemology literally denotes the science of knowledge. It addresses the inquiry of whether knowledge reflects reality or facts, or if it comprises judgements that do not align with facts. Epistemology possesses a rich history extending from the ancient Greeks to contemporary times, with practically every prominent philosopher contributing to its development.

During the study of epistemology, one frequently encounters diverse findings reached by various epistemologists about issues related to knowledge. Certain epistemologists contend that the existence of an object is essential for knowledge. Consequently, all knowledge encompasses both the knower and the known, although some contend that the knowledge possessed by the knower differs from the knowledge of the known, leading to debates about the nature of knowledge and its relationship to reality. Consequently, varying definitions and interpretations of epistemology have resulted in the proliferation of many theories of knowledge. The four principal theories of knowing are rationalism, empiricism, critical theory of knowing, and intuitionism. While several hypotheses exist, we are focusing on the four aforementioned theories, which are the most significant. All these ideas illuminate various facets of knowledge, such as the nature of belief, justification, and the limits of human understanding. The mutual criticism among epistemologists has produced a compelling domain within philosophy.

The history of Western epistemology, derived from the Greek term 'episteme', meaning 'knowledge', encompasses two philosophical schools that seek to define knowledge through various approaches. Rationalism posits that knowledge arises from reasoned thought outside of sensory experience, whereas empiricism contends that knowledge is derived from sensory experiences. Critical philosophy, as articulated by Immanuel Kant, is a philosophical

framework that critiques both rationalism and empiricism simultaneously, subsequently establishing a synthesis between the two.

Rationalism

In rationalist philosophy, reason is regarded as the source of knowledge. The Eleatics, significant figures in Greek philosophy, laid the foundation for prioritising reason over sensory experience as a source of knowledge. Platonism significantly influenced this preference. On the other hand, Descartes, who is often regarded as the founder of contemporary Western philosophy, personifies the essence of rationality. He asserts that the sole method for obtaining genuine knowledge is through reason, which he argues is the mechanism that enables us to distinguish between reality and illusion. Spinoza and Leibniz are regarded as prominent rationalists. Rationalists contend that specific fundamental truths of reality are inherent and acknowledged as true by reason. They assert that these principles are the foundation of reality. Consequently, we assert that rationalism is a philosophy that prioritises unaided reasoning in the acquisition and justification of knowledge.

Some important criticisms levelled against rationalism are as follows:

- The notion of rationalism assigns secondary significance to sensory perception in the acquisition of knowledge. Nonetheless, critics assert that it is erroneous.
- The rationalists do not offer any justification for their belief in the innateness of concepts, nor do they reach a consensus on the number of innate ideas.

Empiricism

Many philosophers contend that empirical experience is the sole source of knowledge. They assert that the sole method of acquiring knowledge is through the sensations derived from one's sensory organs, which means that all knowledge must be based on observable and measurable phenomena rather than intuition or reasoning alone. The term "empiricism" denotes both the concept and the associated attitude. They uphold the conviction that sensory experience constitutes the foundation of all genuine knowledge, and they strive to build a link between knowledge and experience. John Locke, a British philosopher, is a pivotal player in the history of current empiricism. Empiricists dispute the theory of intrinsic conceptions because they

believe all knowledge comes from sensory experience. "Any knowledge acquired by humanity is fundamentally rooted in and reliant upon experience," stated G.T.W. Patrick. "Experience serves as the impetus of existence." Sensation constitutes the foundation of all information, and ultimately, all our thinking can be distilled to sensory experiences. Generalisations based on experience represent the sole means of obtaining high-quality knowledge, as they allow us to draw conclusions and make predictions about the world based on our sensory interactions and observations. The climax of empiricism is the type of scepticism found in David Hume's philosophy. Hume's philosophy is defined by his rejection of the notion that conceptions themselves constitute the objects of our knowledge. One of the several critiques of empiricism is that it relies on empirical data.

- It fails to provide an adequate explanation of perception.
- It fails to provide an adequate explanation of perception.
- It contradicts contemporary psychology.
- It fails to elucidate the relationship between ideas and concepts, as well as the differentiation between knowledge and ideas.
- The assertion that the mind is inactive during the formation of concepts is erroneous, as it is highly engaged in the selection of sensations and the conceptualisation process.

Critical Theory of Knowledge

Prior to Kant's emergence in Western philosophy, the predominant schools were empiricism and rationalism. John Locke and Berkeley contributed to the evolution of the former, while David Hume's philosophical writings were the apex of this intellectual tradition. Descartes originated the latter, Spinoza improved it, and Leibniz elevated it to its apex in his writings. Leibniz contended that all knowledge is a priori, whereas Hume maintained that only empirical evidence can underpin any philosophy. Hume asserted that only mathematics and science were conceivable, rendering philosophy impossible, as there was nothing beyond experience, which undermined Kant's belief in empiricism. His objective is to establish a synthesis between the rationalist and empiricist philosophical traditions. He takes up Leibniz's premise in his well-known work, Critique of Pure Reason, and elucidates his critical theory of knowing. In his perspective, knowledge is neither entirely a priori, as asserted by rationalism, nor completely a posteriori, as posited by empiricism. He contends that knowledge is not either of those concepts. He asserted that both empiricism and rationalism were dogmatic; the former for

accepting the validity of sensations, and the latter for supposing the existence of innate concepts. He contended that rationalism was inherently dogmatic. Consequently, Kant was the inaugural philosopher to offer a conclusive rebuttal to the sceptical conclusions posited by Hume. His critique of empiricism and rationalism demonstrated that both philosophical schools were prejudiced and only valid within a restricted context, which ultimately led to a more comprehensive understanding of knowledge that transcends the limitations of each school. He also illustrated that we might elucidate the process of knowledge by integrating the two philosophical schools, suggesting that a synthesis of empirical observation and rational thought can lead to a more comprehensive understanding of reality. While he recognises that the senses facilitate the acquisition of information, he does not concede that sensation alone is the foundation of human experience and comprehension. Our awareness of mental activities underpins his conviction in the existence of the mind, suggesting that cognitive processes, such as thoughts and emotions, have an important influence on our understanding of reality beyond mere sensory input.

Kant concludes that the acquisition of knowledge allows for the possibility of synthetic a priori judgements, so affirming the viability of philosophy. He asserts that pure conceptions lacking precepts are devoid of significance, whereas pure precepts absent of concepts lack visibility, indicating that both elements are essential for meaningful understanding in philosophy. Kant, through his critical theory of knowledge, argued that neither reason nor sensory experience, when separated, can yield a genuine comprehension of reality.

Intuitionism

Proponents of intuitionism assert that intuition serves as the instrument of knowledge, deeming reason and intellect inadequate for understanding the surrounding universe. Henri Bergson, a French philosopher, has recently emerged as a proponent of intuitionism. He thinks that intuition is the faculty responsible for genuine knowledge, while he criticises intellect for being limited in its ability to grasp the complexities of reality. G.T.W. Patrick asserts, "The sole means of comprehending reality is by intuition." "Intuition reveals a dynamic environment that is ever-evolving and presenting novel experiences." The essence of creativity is entirely absent from intellectual knowledge, and only intuitive experience can furnish a precise understanding of it. Bergson critiques intellect for several reasons, notably its tendency to distort our experience of reality. He defines intellectual knowledge as conceptual knowledge, distinguished by its static, analytical, and relative nature. He defines intellectual knowledge as conceptual knowledge,

distinguished by its static, analytical, and relative nature. Conversely, employing our intuition engenders a feeling of unity with our surrounding environment, which contrasts with the more detached nature of intellectual knowledge, highlighting how intuitive knowledge can lead to a deeper emotional connection and understanding of our experiences. Intuition is a definitive and comprehensive form of knowledge. Due to its dynamic, synthetic, and absolute nature, it offers an internal image of our surrounding reality, which can lead to subjective interpretations that may not align with objective facts, potentially resulting in misunderstandings or misjudgments about the world around us. Bergson's theory of intuitionism is susceptible to critique based on the following points. Philosophy cannot offer a rational comprehension of reality only based on intuition, as it depends on intellectual knowledge and is incapable of delivering such a perspective. Both intellect and intuition are essential, and both must possess the ability to analyse and synthesise information independently. Intuition must culminate in a logical conclusion. Bergson offers a rational argument for the result of his intuition.

Correspondence Theory of Truth

The correspondence theory of truth is occasionally called the "theory of accordance". This theory posits that truth is determined by maintaining a link between "a belief regarding X" and a corresponding fact about "the same X in reality". This idea asserts that truth is determined by maintaining this link. This is what we refer to as a "theory of common sense". We typically assume that any thought is validated as true or incorrect based on its link to the external physical reality, rather than its relation to other beliefs or the beliefs of others. To clarify this perspective, consider the following examples: My assertion that my house contains a king-size bed is only valid if it aligns with a specific fact about my house; conversely, the belief that there is a mountain within my house is invalid, as it does not correspond to that specific fact. Moreover, although we have only employed examples from the external-physical realm, this theory is also pertinent to our perceptions of the internal or experiential world. My genuine experience of such pain is the sole evidence that supports my assertion of having endured a terrible headache two days earlier. A schematic representation of this idea may look like the following:

Identity Theory

Identity Theory, a significant school of thought in the philosophy of mind, aims to provide a scientific elucidation of the attributes of mental states and is also known as the Mind–Brain Identity Theory. This theory posits that mental states are not separate from or reducible to entities beyond the physical realm; instead, they are identical to the physical states experienced

by the brain. This indicates that feelings like pain, pleasure, belief, perception, and emotion are only the result of unique neuronal processes within the neurological system. Mental experiences are not just the result of brain activity; rather, they are brain activity articulated in psychological terminology. The thesis vehemently contests the dualist viewpoint, primarily linked to René Descartes, which asserts that the mind is a non-physical entity apart from the body. The theory unequivocally dismisses this perspective, positing an identity instead. Identity theory contends that the assertion of a non-material mind is both scientifically redundant and philosophically contentious, especially as it provokes challenging enquiries regarding the interaction between a non-physical entity and a physical body. Identity theory posits that all mental experiences can, in principle, be elucidated through physical and neurological processes, analogous to the explanations provided by the physical sciences for other natural phenomena.

Identity theory originated in the mid-twentieth century, a period characterised by swift advancements in neuroscience, psychology, and the philosophy of science. This movement was a component of the broader scientific materialism movement, which highlighted that reality is inherently physical and that scientific methods are the most dependable means of comprehending it. Influenced by advancements in neuroscience and discontent with behaviourism and Cartesian dualism, philosophers like U. T. Place, J. J. C. Smart, and Herbert Feigl contended that mental states ought to be equated with brain states, analogous to how scientific revelations have united seemingly disparate phenomena, such as lightning and electrical discharge, as identical. In this discussion, identity theory represents a substantial endeavour to reconcile the philosophy of the mind with the findings and methodologies of modern science.

Considering the limitations of behaviourism, contemporary materialism, frequently referred to as the identity theory of mind, presents a logical response. This hypothesis posits that the mind and the brain are identical entities. 'One-step ontology' refers to the concept that mental states and processes are synonymous with those of the brain. This issue in behaviourism is resolved as it allows for the attribution of a causal and explanatory role to mental processes, thereby bridging the gap between physical brain activities and subjective experiences. Having elucidated the advantages of this idea, we shall now examine it in greater detail. This theory recognises the logical relationship between mental states and behaviour, allowing one to articulate statements such as, "I drank due to my thirst."

The identity theory posits that the mind can be equated with the brain. In other words, it posits that mental states are identical to the brain's physical states. The categorisation of 'physical' is a crucial characteristic. Ultimately, property dualism asserts that mental states are attributes of the brain. The notion of property dualism posits that mental states are non-physical features of the brain. Consequently, for the identity theory to distinguish itself from property dualism, it must assert that mental states are only physical states of the brain. The identity theory derives its name from its assertion of a relationship between distinct mental states and corresponding physiological states of the brain. The notion that mental states correspond to brain states is not a fresh concept. The French philosopher Pierre Gassendi and the English philosopher Thomas Hobbes, who were contemporaries, posited the claim more than three hundred years ago. Conversely, the concept was not clearly articulated and debated until the 1950s, when a cohort of Australian philosophers, including J. C. Smart, engaged with it.

A collective of philosophers aimed to supplant dualist interpretations of the mind with empirically based explanations, significantly contributing to the formulation of this theory, which arose in the mid-twentieth century as part of a broader shift towards scientific materialism. Key philosophers linked to Identity Theory include U. T. Place, J. J. C. Smart, and Herbert Feigl.

U. T. Place was among the initial advocates of identity theory. In his renowned paper “Is Consciousness a Brain Process?” (1956), Place asserted that consciousness ought to be considered a brain activity rather than a non-physical phenomenon. He contended that mental experiences are not separate from the physical domain but are analogous to cerebral functions. Place likened the mind-brain identification to scientific identities like "lightning is an electrical discharge", implying that seemingly distinct events may actually represent the same object articulated in many ways. His work created the groundwork for later and more systematic versions of identity theory, which further explored the relationship between mental states and brain processes, providing a more comprehensive understanding of how these identities function in various contexts.

Herbert Feigl is another notable philosopher associated with Identity Theory, which posits that mental states are identical to physical states in the brain. Feigl advocated for a kind of identity theory based on logical empiricism and scientific realism. He contended that mental states and physical brain states are nomologically analogous, indicating their connection through natural principles established by scientific investigation. Feigl emphasised that the equivalence between mind and brain is not an obvious or analytic truth but rather an empirical discovery, like the findings in other scientific disciplines. His work facilitated the unification of the

philosophy of mind with the philosophy of science, demonstrating how empirical findings in neuroscience can inform philosophical discussions about consciousness and mental states.
J. J. C. Smart is regarded as the most prominent and influential proponent of Identity Theory. Smart ardently attacked the prevailing Cartesian dualism by asserting that sensations are merely brain processes. He believed that mental and physical descriptions correspond to identical experiences but use different terminology. The expressions "I am in pain" and "my C fibres are firing" both articulate the same experience. Smart also proposed the concept of topic-neutral language, indicating that mental reports do not obligate us to affirm the reality of non-physical entities. Identity theory emerged as a pivotal stance in modern philosophy of mind due to its focus on scientific elucidation and materialism throughout his oeuvre.

The notion was examined by J. C. Smart. Smart employed several metaphors to convey his thesis that mental states are identical to brain states while expressing the identity theory. That is the assertion made by Smart. Just as water is H_2O and lightning is an electrical discharge in the environment, mental states correspond to the states that occur in the brain. The analogies employed by Smart are essential; it is imperative for us to understand the significant distinction between tokens and types. Let us commence with an illustration. The Australian outback is home to numerous sections that harbour dingoes, a species of wild dog. Envision a scenario where we are camping in a remote location and encounter four dingoes roaming about our campfire. Considering this, we own four identical tokens. The individual animals are denoted by the l tokens, while the type pertains to the classification or category of creatures to which the individuals are assigned. Within this discourse, identity theories are categorised into two types: type–type identity theories, which assert that every category of mental state corresponds to a specific category of physical state, and token–token identity theories, which claim that each individual mental state corresponds to a specific individual physical state.
When enquiring about the number of letters in the word 'tree', the accurate response is either three or four, contingent upon whether one considers letter-types or letter-tokens; specifically, there is one token each of the letter-types 't' and 'r', and two tokens of the letter-type 'e'. Currently, in modern philosophy of mind, type-type identity theories have been challenged on the basis that kinds of mental states are, arguably, 'multiply realisable'.
To offer a robust alternative to behaviourism and dualism, Place, Smart, and Feigl collaborated to formulate identity theory. Nonetheless, their perspectives faced further criticism, particularly on multiple realisability and the issue of qualia, which contributed to the emergence of ideas like functionalism, as critics argued that mental states could be realised in various ways across

different systems, challenging the notion of strict identity between mental states and brain states. Notwithstanding these critiques, the contributions of these philosophers remain fundamental.

(B) Behaviourism

It is a theory within the philosophy of mind that elucidates mental concepts through observable conduct and behavioural tendencies rather than internal, secret mental processes. Philosophical behaviourism is alternatively referred to as logical behaviourism. The philosophy of mind constitutes a subdivision of philosophy. The principal aim of this study is not to elucidate the process of behavioural acquisition; instead, it focuses on the significance of mental terminology. This perspective allows the interpretation of statements regarding an individual's thoughts, aspirations, and feelings as predictions about their behaviour in specific situations.

A collective opposition to Cartesian dualism, logical positivism, and ordinary language philosophy influenced the emergence of behaviourism in philosophy in the mid-twentieth century. Dualism posited the mind as a non-physical entity apart from the body, which engendered substantial philosophical dilemmas regarding the mind-body interaction. By reconceptualising mental language through the lens of behaviour, proponents of the philosophical behaviourist school sought to eliminate these problems, arguing that understanding mental states in terms of observable behaviours could provide a more coherent framework for addressing the mind-body interaction issues raised by dualism.

Gilbert Ryle, especially through his work The Concept of Mind (1949), is regarded as the most important philosopher associated with the philosophical behaviourist school of thought. Ryle's characterisation of the mind as a "ghost in the machine" served as a critique of Descartes' viewpoint. He argued that Descartes committed a category error by regarding the mind as an entity of the same logical type as the body. Ryle asserts that the mind is not a substance but rather a means of describing an individual's activities and behaviours.

Behaviourism is a philosophical doctrine that suggests we most effectively comprehend mental processes as behavioural patterns. Observing that an individual tends to carry an umbrella, consult the weather forecast, or refrain from venturing outdoors without protection exemplifies the indication that the person anticipates rain. Mental words do not denote hidden internal events; instead, they signify behavioural patterns, both present and prospective.

Philosophical behaviourism achieves conceptual clarity by denying the existence of unaccounted internal entities, which is one of its primary strengths. It resolves the typical challenge of mind–body interaction by dismissing the notion of a private mind existing

independently of the physical body. Furthermore, it underscores the public and observable characteristics of mental concepts, rendering them amenable to objective observation and analysis, which allows researchers to study mental processes in a systematic and empirical manner.

Philosophical behaviourism aligns well with the scientific spirit, since it prioritises observable behaviours over unverifiable subjective sensations. It challenges the assumption that mental explanations necessitate introspection or private awareness in order to be considered genuine. This action facilitated the shift in philosophy from speculative metaphysics to a more straightforward analytical approach, which emphasises clarity and logical reasoning over abstract theorising.

Regarding the reliability of introspection as a means of acquiring knowledge about our mental states, several philosophers express scepticism, while others even challenge its reality altogether. The philosophical doctrine known as behaviourism most acutely articulates this cynicism. Behaviour, the concept behind the science of behaviour, forms the intellectual basis for the modern field of behaviour analysis, which includes both theoretical and applied research, as well as actual implementations. Behaviourism posits that the interaction between response and environment, including social factors, is the primary determinant of conduct. This pertains to both human and animal behaviours.

The pursuit of a theory of mind and personality to substantiate this novel approach in philosophy was termed behaviourism. The philosophical school of behaviourism asserts that ideas and feelings are not concealed mental states subject to introspection; instead, they are the observable behaviours of evident organisms. Consequently, fear is a tendency to flee; fury is a tendency to annihilate; desire is a tendency to favour; belief is a tendency to agree; and so forth. The conviction that behaviour is fundamental to understanding mental phenomena. The phrase can denote both a philosophical system and a scientific study programme in the realm of psychology. Consequently, we distinguish between scientific behaviourism, grounded in psychological and methodological principles, and philosophical behaviourism, rooted in logical and analytical ideas.

Specifically, Gilbert Ryle was the individual who introduced behaviourism to the realm of philosophy. In "The Concept of Mind", Ryle proposes an innovative approach to the mind-body problem. Reductive materialism is a philosophical perspective that posits that all discourse regarding mental states and processes may be elucidated through a series of assertions about an individual's observable behaviours or predispositions to act. This is the fundamental principle of philosophical behaviourism. Consequently, there is no necessity to

invoke an inner existence or mental phenomena, such as desires, beliefs, moods, or emotions, as if they were autonomous entities.

Behaviouralists assert that the only evidence we possess regarding the mental states of individuals, including our own, is observable behaviour, encompassing both verbal and non-verbal displays.

Behaviouralism emphasises the various behaviours individuals are inclined to display in reaction to different conditions. Philosophical behaviourism, commonly known as "logical" or "analytic" behaviourism, offers a physicalist interpretation of the concept of "mental states" in response to the inquiry, "What are mental states?" Mental states are tendencies to exhibit specific behaviours under particular conditions, as posited by the philosophical perspective of behaviourism.

The philosophical idea of behaviourism posits that experiencing pain necessitates a predisposition to exhibit specific behaviours in response to certain events. The inclination to weep, flinch, or moan exemplifies the manifestations of pain, particularly following an accident such as a fractured leg, a scorched hand, or a comparable affliction. The initial cluster of dots signifies that the behaviours associated with pain extend beyond merely sobbing and wincing; indeed, numerous other actions occur when individuals endure severe pain. Similarly, the second group of dots aims to illustrate that the situations associated with pain extend beyond broken legs and burnt hands; in fact, several stimuli, such as emotional distress, chronic illness, and environmental factors, can induce pain.

Behaviourism, as a philosophical doctrine, does not claim that our mental states are the determinants of our tendencies to respond in specific ways when faced with certain circumstances. According to the philosophical behaviourist perspective, pain does not create our desire to behave in specific ways in response to bodily events; instead, it is a predisposition to react in particular manners when certain occurrences affect our bodies. Philosophers aligned with the behaviourist school of thought use the term "behaviour" to refer to the manifestation of physical occurrences. The body's physical responses encompass weeping, flinching, escaping, extending, and shouting. Additional physiological reactions encompass weeping and flinching. Similarly, behaviourists focus solely on the physical factors that elicit behaviour. This leads to the conclusion that philosophical behaviourism offers a physicalistic interpretation of mental states. For instance, fracturing a leg or incurring a burn on a hand are examples of physical events. According to the philosophical doctrine of behaviourism, mental

states are characterised as tendencies to act in certain ways under specific circumstances. Both the behaviour and the factors that precipitate it are considered physical phenomena.

Radical behaviourism is a distinct scientific theory that must be assessed based on established scientific standards to determine its efficacy in predicting and elucidating the phenomena it aims to address. Edward Thorndike (1874–1949), John Watson (1878–1958), and Ivan Pavlov (1849–1936) were pivotal figures in the evolution of radical behaviourism. The antiquated empiricist notions of the mind provide the foundation of this philosophical school. These beliefs assert that the mind is a tabula rasa, or blank slate, at birth, upon which experience generates sensory perceptions. A process of association that accurately reflects the occurrence of events in reality underlies the development of complex conceptions, which means that as individuals encounter various experiences, they form connections between these experiences that lead to a deeper understanding of the world around them. These concepts are derived from experiences and subsequently amalgamated to generate more intricate notions, which can include abstract ideas formed through repeated interactions and reflections on those experiences. Their recommendation was to examine not the connections between conceptions, but rather those that are physically defined.

This perspective is maintained by "scientific" behaviourists, who assert that the study of mental states, which scientific psychology partially purports to be, should exclusively depend on objective empirical evidence verified by several independent observers. Conversely, introspection is fundamentally a private and subjective matter.

Radical behaviourists have a unique perspective on human purpose and motivation. It is believed that a history of consequences for specific conduct governs the behaviour identified as "operant". A person does not "mean" to activate a light switch; instead, they engage the switch based on prior experiences that have consistently produced light.

In the specific reductionist framework termed "analytical behaviourism", it was posited that mental events may be correlated with tendencies toward observable behaviour. Ultimately, analytical behaviourism is merely a semantic or philosophical proposition, assessed by its efficacy in encapsulating the mental constructs it purports to examine.

Moreover, there exists a third perspective termed "methodological behaviourism", which posits that the sole evidence for any mental experience must be of a behavioural type. This assertion pertains to evidence that is really independent of the other two viewpoints, notwithstanding its frequent association with them. From the standpoint of a methodical behaviourist, psychology

is confined to the pursuit of laws that define the relationship between stimuli and behaviour. When we discuss "stimuli", we denote both the sensory inputs the organism currently perceives and any relevant sensory inputs previously encountered. Let us examine a concise example.

A cage is provided with a light, a lever, and a rat placed within it. One of the dietary pellets. The animal will be let into the cage solely if the bar is pressed subsequent to the activation of the light. While exploring the enclosure, the rat erroneously activates the lever when the light is illuminated, leading to the procurement of a food pellet. Within a short period, the rat's behaviour is modified to consistently activate the lever whenever the light is illuminated. When the light is illuminated, we often assert that the rat has learned to obtain food by pressing the lever. However, this is no longer the situation. In the subsequent paragraphs, we shall see that the methodical behaviourist will be hesitant to use conventional psychological language, such as "learn".

This exemplifies the rule of effect: when an organism receives a reinforcer shortly after exhibiting the operant in reaction to a stimulus, its propensity to produce the operant in response to that stimulus will increase. The law of effect serves as a pertinent example. The methodical behaviourists identify this as the quintessential illustration of the law they seek, as it delineates a relationship between stimulus and behaviour.

Methodological behaviourism instructs psychologists to ignore internal mental states and concentrate on studying behavioural responses to various stimuli. The objective of this research is to identify the laws that connect stimulation with behaviour.

They contend that attributing a mental state to an individual is merely assigning that person an appropriate behavioural propensity. This holds true irrespective of the context in which the attribution occurs. In this debate, "behavioural disposition" denotes an individual's tendency to act in a specific way when faced with certain events and situations.

A rational behaviourist might assert that attributing to John a belief that it is raining only involves assigning to him a propensity to engage in actions such as taking an umbrella while departing from home. This would provide a rational interpretation of the circumstances. This description aims to encompass both "attitudinal" moods and sensory sensations, including pain and nausea. Attributing a sensation of pain to his big toe implies that he may exhibit behaviours such as wincing and groaning at contact, limping when required to walk, and declaring discomfort if asked about the toe's condition, among others.

Arguments in favour of behaviourism

This constitutes the preliminary argument. An individual demonstrates a specific behavioural inclination when desiring coffee, namely, a propensity to consume it. If an individual consistently expresses a wish for coffee yet continually refuses it when presented, it is logical to infer that they do not genuinely desire coffee. This study elucidates a significant aspect of mental states, indicating a robust correlation between mental states and the propensity to exhibit behaviours characteristic of those states. The significant correlation indicates that an individual's inconsistent display of the behavioural traits characteristic of a specific mental state M constitutes persuasive evidence that the individual is not experiencing mental state M. If the philosophical behaviourist's argument that the desire for coffee equates to the inclination to consume it is accurate, it follows that an individual who desires coffee is predisposed to drink it. The philosophical behaviourist's examination of mental states explicitly accounts for the connection between mental states and behavioural tendencies. The strong relationship indicates that an individual's failure to consistently display the behavioural traits characteristic of a specific mental state M serves as compelling evidence that the individual is outside of that mental state M.

A robust correlation exists between mental states and individual behaviours. Philosophical behaviourism, by positing that mental states are only behavioural tendencies, effectively elucidates the link. The association between mental states and behaviour supports the assertion that the philosophical behaviourist perspective is valid.

Verificationism refers to the concept advocated by the Vienna Circle for comprehending the significance of claims. The significance of each assertion is contingent upon the verification method employed. One can ascertain the veracity of a statement through sensory observations, encompassing visual, auditory, and tactile sensations.

The cat is currently situated on the carpet. Evidence for the accusation can be established by locating the cat, confirming its existence, or perceiving its sounds. The statement 'The cat is on the mat' posits that, according to the verificationist theory, a typical observer, upon accurate observation, will visually perceive a cat on a mat, tactilely discern a cat on a mat, and audibly detect sounds associated with a cat on a mat. Verificationism posits that the meaning of a statement is determined by its verification method. What methodologies can be employed to substantiate the assertion that "Bloggs is in pain"? We wish to inform you that Bloggs exhibits sobbing or wincing in reaction to certain biological activities. The verificationism theory asserts that the significance of the statement "Bloggs is in pain" relies on the assumption that a typical observer, in suitable conditions and in reaction to specific events affecting Bloggs's body, will either hear Bloggs crying or see Bloggs wincing, among other potential

observations. If the statement "Bloggs is in pain" is intended to express this meaning, then pain must be the behavioural tendency to cry or wince in reaction to certain bodily events.

Arguments Against Philosophical Behaviourism

The initial discourse. Envision that Bloggs has resolved to embody the quintessential essence of resilience. Despite having stubbed his toe, he exhibits no emotional response, such as blinking, sobbing, or touching the injured area; instead, he proceeds as if nothing occurred. Despite having shattered his leg, he refrained from screaming or crying; instead, he calmly proceeded to the next hospital. Bloggs undoubtedly continues to experience pain; stubbing his toe or fracturing his leg remains painful for him, but he is no longer prone to crying, flinching, or displaying comparable emotional responses. Conversely, each member of that group abstains from exhibiting any behaviours indicative of discomfort, such as weeping, flinching, or vocalising distress. The illustration demonstrates that the expression of unpleasant behaviour is not a requisite for the sense of suffering.

Hilary Putnam, a modern American philosopher, was the inaugural individual to articulate this idea. He coined the term "super stoics" to characterise individuals such as our resolute acquaintance, Bloggs. The exemplary stoic demonstrates that experiencing pain does not necessarily result in displaying pain-related behaviour. An identical example illustrates that exhibiting pain behaviour does not inherently guarantee the sensation of pain, as evidenced by those who are pain-free. They wouldn't feel pain if they stubbed their toe or broke their leg. He develops the capacity to feign profound suffering. When he accidentally stubbed his toe, he recalls saying "ouch" and adjusting the affected area. After some effort, he can now exhibit pain behaviours that are indistinguishable from those of the average individual. Nevertheless, he does not encounter any hardship; he epitomises a "perfect imposter".

The examples of the perfect pretender and the super stoic demonstrate that exhibiting pain behaviour is neither necessary nor sufficient for actually experiencing pain. A tendency to display unacceptable conduct is not a need for the experience of pain. Conversely, one may endure pain without manifesting any behavioural indicators of distress, and one may also be predisposed to demonstrate pain behaviours without genuinely suffering pain. Consequently, the experience of pain does not create a predisposition to respond in specific manners when faced with certain circumstances.

This constitutes a secondary argument. Behaviourism is a philosophical doctrine asserting that each mental state correlates with a specific set of behaviours. Behaviours such as weeping, grimacing, shouting, or contacting the affected region exemplify responses that may occur when an individual is in pain. Should you suspect the presence of a lion nearby, you are likely

to engage in one or more of the following actions: withdraw to the vehicle, retrieve your firearm, or summon assistance. Envision a lion in proximity, and you believe that the optimal strategy to evade an attack is to remain completely motionless. In that scenario, you would refrain from returning to the truck or attempting to recover your firearm; instead, you would remain completely still and motionless. The examples provided demonstrate that our reactions to circumstances are contingent upon the concepts and goals we aim to accomplish. A superstoic's response to pain is influenced by the nature of the discomfort and his inclination to convey that he is not in distress. Your preconceived beliefs of lions will influence your response when encountering an approaching lion. I am disillusioned with my life and have a profound urge to cease my existence. Consequently, I shall relocate, as I am convinced that inaction will lead to my demise, and I harbour a yearning for self-destruction. This picture demonstrates the profound link between mental states and behaviour. Your behaviour is predominantly driven by a convergence of multiple mental states occurring concurrently, rather than a singular mental state. Consequently, the philosophical behaviourist method is certain to fail.

Functionalism

Functionalism, which arose in the mid-twentieth century as a counter to behaviourism and the mind-brain identity theory, is regarded as one of the most significant theories of the philosophy of mind. Functionalists define mental states by the roles they fulfil within a system, rather than by their physical composition or the behaviour of the observer alone. Understanding a mental state can be achieved by examining its actions, including the stimuli that elicit it, its interactions with other mental states, and its resultant behavioural outputs. A multitude of philosophers significantly contributed to the evolution and enhancement of functionalism, therefore impacting contemporary discussions in philosophy, cognitive science, and artificial intelligence through their insights.

Hilary Putnam is regarded as one of the seminal figures of functionalism. In the 1960s, Putnam introduced the notion of machine-state functionalism, likening the human mind to a computer program and the brain to hardware. Putnam asserts that mental states resemble computational states in that they are defined by their functional relations rather than their physical constituents. The concept of multiple reality posits that identical mental states can be instantiated in many physical systems, whether human brains, animal nervous systems, or artificial devices, which represents his most significant contribution. This approach directly

challenged the identity theory, which claimed that mental states are equivalent to specific brain states. Putnam's work was crucial in establishing functionalism as a robust and versatile explanation of the mind.

Jerry A. Fodor is a notable functionalist philosopher who integrated functionalist philosophy with a robust dedication to mental representation inside a cohesive philosophical framework. Fodor formulated the Language of Thought (LOT) hypothesis. This theory posits that cognition occurs in an internal symbolic language governed by syntactic rules. Fodor posits that mental states are functional states encompassing representations that hold semantic information. His specific understanding of functionalism emphasised the computational dimension of cognition, which significantly influenced the domain of cognitive research. Fodor argued that functional explanations are crucial for understanding how mental states influence behaviour; hence, they enhance the explanatory efficacy of functionalism.

David Lewis is recognised for formulating an approach to psychology termed 'analytic functionalism'. This method defines mental states based on the causal roles they fulfil, as articulated by conventional psychology. According to Lewis, concepts such as "belief", "desire", and "pain" are delineated by their role within a network of relationships linking perception, action, and other cognitive processes. Once we delineate these functions, we can determine the states that fulfil them, along with their corresponding physical states. Lewis's method sought to reconcile functionalism and physicalism by illustrating that mental notions can be functionally articulated without negating their physical embodiment.

Daniel Dennett, through his concept of an intentional stance, offered a pragmatic and compelling defence of the philosophical school of functionalism. Dennett asserted that an effective approach for predicting a system's conduct is to ascribe beliefs, desires, and intentions to it. This holds true irrespective of the existence of these states as internal, private entities. This perspective posits that mental states are discernible patterns at the functional level. Dennett's work contradicts traditional concepts of consciousness and mentality by broadening the functionalist framework to encompass animals, computers, and artificial intelligence, suggesting that these entities can exhibit mental states similar to humans under certain conditions.

Sydney Shoemaker, who concentrated on mental causality and self-knowledge within a functionalist paradigm, is another notable addition to this domain. Shoemaker asserted that

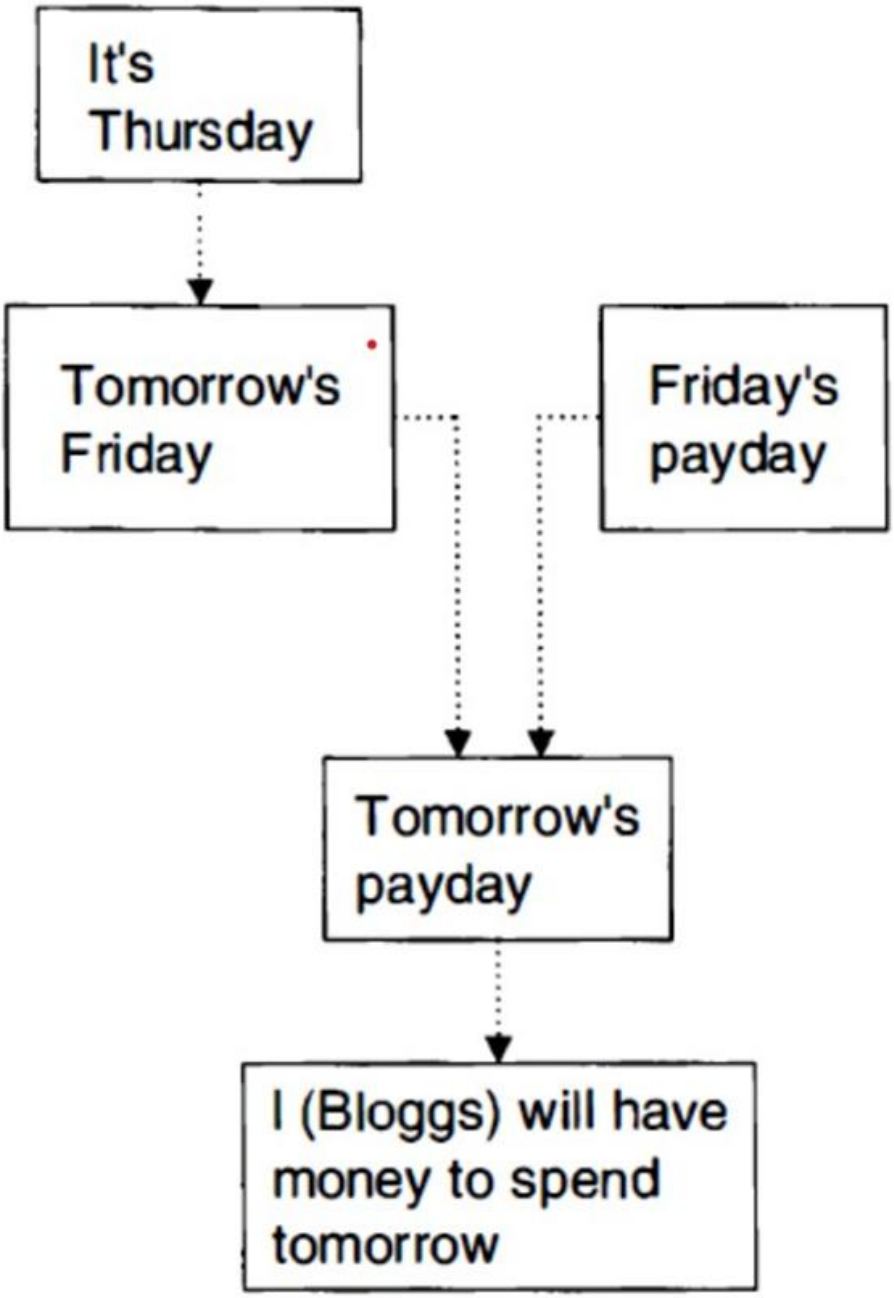
mental states must possess genuine causal powers, and functionalism may explain this without diminishing mental attributes to mere behavioural tendencies. He contended that this aligns with the objectives of functionalism. His work enhanced the conceptual underpinnings of functionalism, enabling him to tackle criticisms regarding its neglect of subjective experience.

From the functionalist philosophical viewpoint, mental states are characterised by their causal interactions with each other, as well as with sensory inputs and behavioural outputs. Functionalist theory is a significant intellectual advancement in analytic philosophy during the twentieth century. It also constitutes the conceptual foundation for a significant body of study in cognitive science. The notion that numerous entities are defined not by their composition but by the functions they execute serves as the foundation for the functionalist perspective in philosophy. The knife's cutting ability, rather than its material composition, separates it from other blades; the difference between steel and plastic is inconsequential to its classification as a knife. The functionalist argues that a comparable premise applies to mental states and processes. The nature of a specific mental state is defined not by its physical composition, such as brain activity, but by the functions it encompasses and the types of causes and effects it produces. The functionalism theory posits that mental states are defined by the functions they generally perform, sometimes referred to as their "functional" duties. A comprehensive functionalist maintains that the causal function of a mental state (M) is the definitive factor for all attributes of M. Consequently, the content of M is determined by the causal function it exhibits. A functional role theory of content elucidates the mechanism by which causal roles are established. The assessment of functionalist role theories of content is often conducted in this manner. In the majority of instances, advance. It is important to recognise that an individual's mental states generally establish a causal network. For instance, proceed in the traditional fashion. For instance, Bloggs's conviction that it is Thursday prompted him to deduce that tomorrow is Friday; his belief that tomorrow is Friday, coupled with his understanding that Friday is payday, culminated in his assumption that tomorrow is payday; and his belief that tomorrow is payday led him to conclude that he will have money to spend tomorrow.

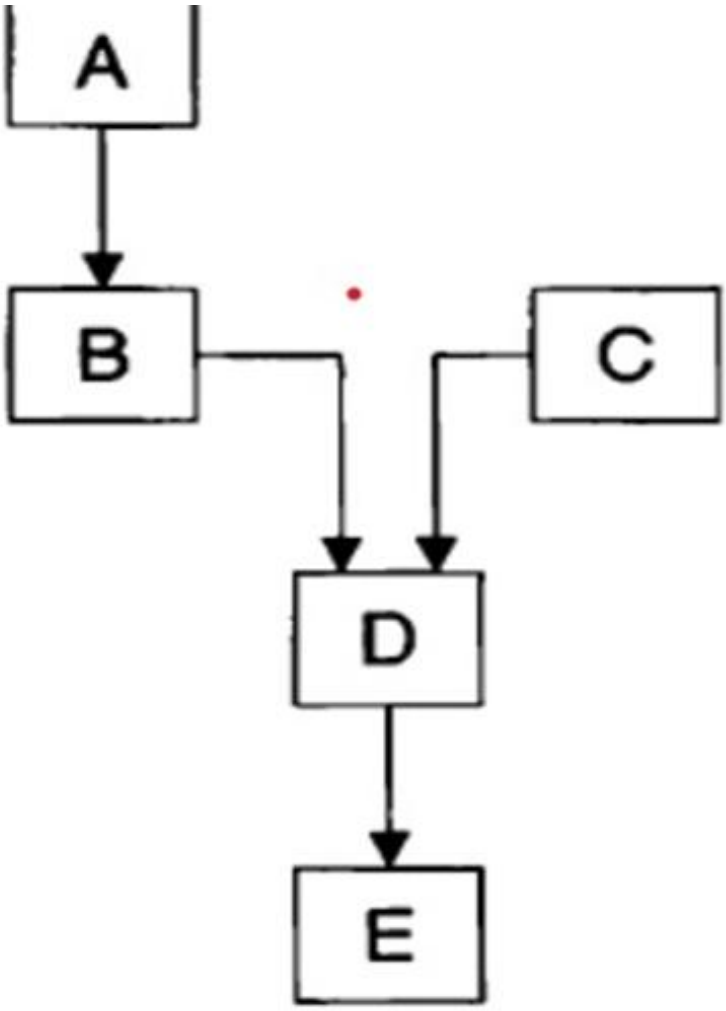
In addition to the causal network that includes mental states, there exists an inferential network that constitutes the essence of those beliefs. Thus far, we have recognised the presence of two networks: a causal network linking mental states and an inferential network correlating the contents of mental states. The essence of Bloggs' ideas in the example is articulated as a logical

argument, advancing from a sequence of premises to the conclusion that "I (Bloggs) will only need to spend tomorrow." Proponents of functional role theories of content believe that individuals ascribe content to mental states by identifying a correlation between the inferential network and the causal network, which suggests that understanding these correlations is essential for comprehending how mental states influence behaviour and decision-making.

An inferential network



A causal network involving five beliefs (A-E)



In humans, the perception of pain is associated with the activation of C-fibres, whereas in squids, it is indicated by the activation of D-fibres. C-fibre activation is believed to fulfil a similar role in humans as D-fibre activity does in Flipper, as per the functionalist hypothesis of

the brain. This perspective posits that the feeling of pain is defined as an internal condition that fulfils a specific function. The 'pain function' is activated by an internal condition resulting from a body injury, prompting us to exclaim 'ouch' and tend to the damaged area. The functionalist view posits that pain's experience results from an internal state induced by physical harm. This state compels us to exclaim "ouch" and manipulate the affected area. The idea of functionalism posits that for an individual to possess mental state M, there must exist an internal condition that satisfies the 'M-function'.

Antibiotics are chemicals that eliminate pathogenic microorganisms while minimising their harm to patients. Medical specialists regard penicillin as an antibiotic because it can eliminate hazardous microorganisms with minimal harm to the patient. Erythromycin is classified as an antibiotic due to its capacity to eradicate harmful germs with minimal harm to the patient. Penicillin and erythromycin possess markedly dissimilar chemical structures.

Functionalism is a philosophical system that posits mental processes are analogous to carburettors and antibiotics in some significant aspects. In contrast to an antibiotic, a carburettor is defined by its function of performing "carburettor work", while an antibiotic is characterised by its role in executing the "antibiotic task". The execution of functions associated with a specific mental state, such as decision-making or emotional regulation, is the defining characteristic of that state.

The notion of diverse realisations asserts that pain can exhibit multiple expressions. From the functionalist perspective, the many forms of pain are as follows. A situation that assumes the function of pain is termed "functionalism" by philosophers of mind. This hypothesis posits that an organism experiences pain if it has an internal state that serves the function of pain or if it possesses a condition that takes on the role of pain.

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The defining elements of the pain role are its inputs, outputs, and internal links. Walking on a tack, fracturing a leg, and scalding a hand are all instances of circumstances that may induce pain. The inputs are the factors that induce pain. The actions elicited by pain are termed 'outputs'. Examples of outputs encompass vocalisations such as shouting, tactile stimulation of the affected area, and uttering "ouch". The internal connections denote pain and other mental states, serving as the causal links between the two. For instance, they encompass the causal link between pain and anxiety, indicating that acute discomfort generally induces anxiety. Walking on a tack may induce discomfort, typically eliciting feelings of dread and exclamations such as "ouch".

Functionalism recognises the impossibility of directly correlating various mental states with distinct behavioural dispositions; yet, it seeks to characterise mental states through behaviour, although indirectly. The functionalist hypothesis posits that the causative role of a mental state can be delineated into three distinct types of interactions deemed causal.
The way light entering the eyes creates a visual impression exemplifies how specific elements of an environment might first evoke distinct mental processes. Secondly, specific mental states can causally interact with other mental states within the same individual. This is evidenced by the notion that a painful sensation in the leg may create the impression of an injury to the limb.

Certain mental states can causally influence an individual's bodily actions. An instance of this is when an individual perceives an injury to a leg and thereafter harbours a profound urge to alleviate the resultant pain. This may lead the individual to contact the injured region or withdraw their leg from immediate peril. According to functionalists, the categorisation of a mental state is not fundamentally reliant on its intrinsic material composition; instead, it is determined by the state's functional role within the associated system.

The notion of functionalism aligns with the premise that people and groups may exhibit enough resemblance in their functional structure to share analogous mental processes. The functionalist hypothesis posits that mental states serve as conduits for various causal roles, as demonstrated in our current comprehension of the theory. In essence, a wide range of states can fulfil the functions associated with certain mental states. This allows functionalism to illuminate the diverse realisability of mental states. We shall examine the relationship between functionalism and the following theories: (i) (i) (i) (i) (i) type identity theory; (ii) restricted type identity theory; and (iii) token identity theory.
Per functionalist theory, an organism experiences pain due to a state that fulfils the function of pain perception. Currently, it is a plausible possibility that the function of pain may be executed by a non-physical state of matter. Consequently, it is a plausible assertion that functionalist substance dualism is accurate. They believe it is highly probable that the unique causal roles of various mental states are realised by the brain's physical properties. If functionalism is accurate, a variant of the identity theory is likely to be authentic. This is due to the close relationship between the two hypotheses.

Functionalism and Brain State

The specific functions of the brain can be used to characterise its state. An individual is deemed to be in an adverse cerebral condition if they are undergoing pain. A functionalist framework allows for the expression of numerous forms of realisability. For instance, functional roles might characterise brain states like hunger or pain. Kim shows the concept of diverse realism in functional roles through the use of a mousetrap. In certain instances, a mousetrap may feature a basic trigger mechanism, whilst others may incorporate a wire cage that closes upon sensor activation. Lasers or hydraulic power may be incorporated, contingent upon the circumstances, to enhance the effectiveness and efficiency of the mousetrap in capturing mice. Any device capable of ensnaring a mouse is classified as a mousetrap, irrespective of its dimensions, form, or operation. The identity theory posits a link between mental states and brain states. Each occurrence of the category "pain" is associated with a singular instance of "C-fibre firing concurrently". The activation of C-fibres and the experience of pain possess a specific identity. The identity theory posits that each type of mental state is intimately correlated with a certain neurological condition. Conversely, the identity hypothesis does not require that every brain condition correspond to a specific mental state. The states under consideration are the real ones of the brain. "Moreover, identities should not be determined solely by observation or by examining the meanings of pertinent phrases."

Thomas's belief that it is raining aligns exactly with the cerebral state often elicited by the visual and auditory cues of rainfall. This brain state usually coexists with other states that facilitate the activity of retrieving an umbrella, among other acts. Lewis contended that mental states correspond to the physical states within us and that they have certain causal purposes. This data suggests that the link between specific mental states and specific physical conditions may differ among individuals within the same population. Lewis asserts that the physical condition associated with a specific mental state depends on the causal functions that the state generally fulfils for the entity that possesses it. The perception of pain constitutes a unique brain state observed in humans.

Functionalism and Six Features of Mental State

1. Certain states of the world are the root cause of certain mental states. States of the brain are likely to occupy the functional roles associated with different mental states, as we have seen. Therefore, the assertion that certain mental states are caused by states of the world is only true, according to the functionalist, if certain brain states are caused by states of the world.

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2. Certain mental states prompt certain behaviours. Once more, it is important to keep in mind that it is highly probable that the states of the brain are the ones that occupy the functional

functions that are distinctive of mental states. Therefore, according to the functionalist, the assertion that certain mental states are responsible for actions is only valid if certain brain states are responsible for those acts.

3. Certain states of mind are the root cause of other mental states. Considering that mental states are states of the brain, the assertion that certain mental states are responsible for other mental states is equivalent to asserting that certain brain states are responsible for other brain states. And it is without a doubt the case. Not only is it true that certain mental states are the cause of other mental states, but it is also true that the causal ties between mental states can occasionally mirror the rational relations that exist between them.

4. It is possible to be cognisant of some mental states. Consciousness poses a significant challenge, as it always does. It would appear that it is possible to conceive of a robot that possesses a central computer that is capable of occupying the functional role that is characteristic of pain but does not experience it. If what you say is accurate, then consciousness poses a significant challenge to the functionalist approach.

5. There are mental states that are associated with things or occurrences in the world. Regarding this matter, there are two different theoretical perspectives. There is one of them that is quite comfortable with functionalism, and that is functional role semantics. On the other hand, the other approach, which incorporates the causal theory of content, necessitates additions to the fundamental functionalist framework and, at the very least, some modifications.

6. Many different types of mental states have been found to have a consistent correlation with various types of brain states. It has been brought to our attention that it is quite conceivable that the functional functions that are characteristic of the various mental states are occupied by brain states. As a result, the argument that there are systematic relationships between mental states and certain brain states is compatible with the functionalist perspective.

The ideas of Identity Theory, Behaviourism, and Functionalism offer significant philosophical insights into the nature of the mind and its relationship with artificial intelligence. Each hypothesis provides a distinct elucidation of the emergence of mental states and the

comprehension of intelligent behaviour. Consequently, these philosophical theories facilitate an examination of whether computers may genuinely think, comprehend, or possess consciousness. Behaviourism concentrates on behaviour, Identity Theory underscores biological processes, and Functionalism accentuates functional roles. Collectively, they offer a more profound framework for examining the interplay between mind, cognition, and artificial intelligence, enhancing current discussions regarding the essence of intelligence and the future of AI.

The status of machine knowledge represents a crucial epistemological conclusion derived from artificial intelligence. The common notion of knowledge is that it encompasses belief, truth, and justification as a human endeavour. Artificial intelligence systems may retain knowledge, infer conclusions, learn from data, and provide predictions. This raises the question of whether artificial intelligence systems possess knowledge or merely manipulate symbols and data. Proponents of a stringent epistemological viewpoint argue that artificial intelligence (AI) lacks understanding and consciousness, hence possessing only derived or instrumental knowledge rather than knowledge in the comprehensive human sense.

Furthermore, artificial intelligence transforms controversies regarding the sources of knowledge. Classical epistemology distinguishes between a priori knowledge, which is independent of experience, and a posteriori knowledge, which is derived from experience. Empiricist models of knowledge acquisition resemble machine learning systems since they predominantly depend on data and experience. A priori, conversely, pertains to the characteristics of the algorithms, logical structures, and mathematical models that inform artificial intelligence. Artificial intelligence (AI) obscures conventional epistemic boundaries by integrating elements of rationalist and empiricist methodologies.

Another significant implication pertains to the credibility and elucidation of the statement. Reasons, evidence, and critical reflection are the methods via which human epistemology endeavours to validate perspectives. In the realm of deep learning models, artificial intelligence systems often produce outputs that are very accurate yet challenging to elucidate. Epistemic opacity is an issue that emerges from this, wherein even developers cannot fully ascertain how an artificial intelligence reached a judgement. This raises problems about the epistemic justification of beliefs or recommendations generated by artificial intelligence.

Furthermore, artificial intelligence engenders novel forms of epistemic authority and trust. AI systems are exerting a progressively substantial influence on decision-making across various

industries, including medicine, law, economics, and research. This shifts epistemic responsibility from individual human knowers to the intricate socio-technical systems surrounding them. Epistemologists must now investigate who is responsible for the knowledge claims produced by artificial intelligence: the system, the programmer, the data, or the user, and how trust should be distributed.

The advancement of artificial intelligence further influences the parameters of knowledge. Artificial intelligence possesses the capacity to analyse vast datasets that exceed human cognitive powers, facilitating the identification of novel discoveries and patterns imperceptible to humans. Artificial intelligence is, nonetheless, limited by issues including data quality, biases, and algorithmic architecture. These limitations underscore that the knowledge underpinning artificial intelligence is not objective nor infallible; instead, it is shaped by human ideals and assumptions embedded in its creation.

Artificial intelligence has implications for social and collective epistemology. Collaborations between humans and robots are increasingly contributing to knowledge generation. The importance of concepts such as distributed cognition, expanded mind, and common knowledge is starting to resurge in popularity. The integration of artificial intelligence into the epistemic landscape transforms how societies produce, authenticate, and disseminate knowledge.

The scepticism of identity theory regarding powerful artificial intelligence presents a crucial significance for the field of artificial intelligence. If specific mental states correspond to particular biological brain states, then computers composed of silicon, circuitry, and code cannot genuinely possess mental states. This is due to the inability of computers to cultivate their own cognitive states. The emulation of intellect or consciousness by artificial intelligence systems is feasible; nonetheless, it is crucial to recognise that these systems do not authentically think or experience emotions. This raises doubts about the possibility of artificial intelligence possessing genuine feelings, experiences, or opinions.

Moreover, identity theory significantly emphasises the biological foundations of cognitive processes. This posits that achieving genuine intelligence in artificial intelligence research necessitates the creation of systems that can closely replicate the physical architecture and functions of the human brain. This is essential for attaining authentic wisdom. In this context, simplistic computing models or abstract algorithms would be inadequate. Moreover, identity theory provides conceptual endorsement for several methodologies, including artificial

intelligence informed by neurological processes, brain emulation, and physiologically plausible neural modelling.

the constraint it places on the notion of multiple realisability via its execution. Unlike functionalism, which permits mental states to be instantiated in various physical systems, identity theory confines mental states to specific subtypes of physical states, particularly those linked to the human brain. Consequently, identity theorists may contend that artificial intelligence lacks genuine consciousness due to its dissimilar physical identity compared to the human brain, yet exhibiting indistinguishable behaviour from a human consistently. This is due to the fact that artificial intelligence lacks the identical physical characteristics of a human brain.

Conversely, identity theory does not diminish the practical applications of artificial intelligence in real-world contexts. From this vantage point, artificial intelligence (AI) remains a valuable instrument for emulating human cognition, predicting behaviour, and assisting in decision-making. These are all domains where AI can be profoundly advantageous. Artificial intelligence models can provide neuroscientists with testable simulations, significantly aiding their understanding of brain function. Although the models lack consciousness, this is true.

Behaviourism significantly influenced the early notion of artificial intelligence (AI) by positing that intelligence is associated with behaviour rather than the internal mental processes of an individual. This was because behaviourism argued that intelligence is associated with behaviour. The behaviourist perspective posits that an artificial system can be deemed intelligent if it demonstrates behaviours indistinguishable from those of a human. This is due to the fact that the conduct of an artificial system is indistinguishable from that of a human being. The Turing Test, devised by Alan Turing, evaluates computer intelligence exclusively by observable linguistic behaviour, disregarding the machines' internal mental states. This concept is closely related to the Turing Test, which evaluates computer intelligence. Consequently, behaviourism offered a theoretical rationale for assessing artificial intelligence only based on its performance, rather than its underlying design.

In artificial intelligence, behaviourism is defined by its significant emphasis on the relationships between input and output. During the initial phases of artificial intelligence and machine learning development, the stimulus–response paradigm from behaviourist psychology was considered. These systems were designed to provide suitable outputs in reaction to specific

inputs. Reinforcement learning is a type of learning that embodies behaviourist principles. This form of learning is defined by the application of rewards and penalties to influence an agent's conduct. In this regard, behaviourism significantly contributed to establishing the foundation for the advancement of realistic artificial intelligence systems that learn from their environment through feedback.

Behaviourism posits that internal mental states are not requisite for the existence of intelligence. An artificial intelligence system need not possess consciousness, understanding, or subjective experience to be deemed intelligent; it suffices for it to exhibit behaviour consistent with intelligent conduct. Weak artificial intelligence refers to a form of artificial intelligence where machines emulate intelligent behaviour without asserting genuine understanding or cognition. This facilitates the development of weak AI, a specific category of artificial intelligence. Consequently, behaviourism refrains from participating in intricate metaphysical discussions over the existence of mind and consciousness in machines.

The behaviourist approach to artificial intelligence has repeatedly faced significant criticism. The assertion that participation in intellectual endeavours does not inherently signify genuine understanding is a significant worry that arises. John Searle's Chinese Room argument challenges behaviourism by illustrating that a system can generate accurate responses without comprehending their meaning. Some individuals, however, contend that behaviourism neglects interior cognitive processes like as representation, thinking, and intentionality. These procedures are crucial for elucidating higher-level intelligence.

The ethics of artificial intelligence also have a significant impact. Artificial intelligence systems lack the capacity for interior sensations such as joy or misery, and hence, they cannot perceive moral harm. This is predicated on the idea that behaviourism is valid. Consequently, artificial intelligence systems lack a moral stance and bear full responsibility for ethical conduct. The onus for ethical conduct is with individuals. While this facilitates ethical decision-making, it may pose challenges in the future if artificial intelligence systems exhibit increasingly human-like behaviours.

In conclusion, behaviourism has significantly influenced the development of artificial intelligence and the evaluation of its efficacy. The emphasis of behaviourism on observable behaviour, performance, and learning through reward accounts for this phenomenon. It was a significant paradigm for early artificial intelligence research; but, its failure to account for the mental processes occurring within the mind limits its ability to accurately depict true

knowledge and consciousness. This is notwithstanding its status as a beneficial and significant paradigm. Despite the emergence of more intricate cognitive and functional theories, behaviourism remains a significant historical and philosophical foundation for the ongoing development of artificial intelligence (AI). This is the case despite the fact that these theories have taken their position.

Functionalism has been a significant and influential aspect in both the research and application of Artificial Intelligence (AI). Functionalism offers a substantial conceptual basis for artificial intelligence (AI) by promoting the notion of multiple realisability. This strategy enables the attainment of an identical mental state across diverse physical systems. Examples of these physical systems encompass the human brain, the neurological systems of other animals, and artificial robots. Consequently, researchers in the field of artificial intelligence need not replicate the biological architecture of the human brain to develop intelligent systems. Conversely, these individuals concentrate on emulating the functional organization of cognition, fully aligned with the functionalist philosophical viewpoint.

Symbolic or classical artificial intelligence serves as one of the earliest implementations of functionalism within the domain of artificial intelligence. The initial models of artificial intelligence were grounded in machine-state functionalism, which constituted their principal source of inspiration. These beliefs propose that the mind functions as a computer system, with mental activities regarded as computational states. Artificial intelligence systems employed symbols, rules, and algorithms to convey their beliefs, objectives, and intentions to the public. Alongside these symbolic representations, cognitive activity like as reasoning, strategising, and problem-solving emerged during the developmental process. This was executed to illustrate the functionalist perspective that mental states are dictated by the causal roles they fulfil within a system. This was executed to highlight their importance.

The emergence of cognitive architectures like SOAR (State, Operator, and Result) and ACT-R (Adaptive Control of Thought—Rational) demonstrates the active presence of functionalism. These systems aim to replicate human cognition by deconstructing the mind into its distinct functional components. The components encompass perception, memory, learning, and action, among others. A component is defined not by its physical manifestation but by the function it performs. This similarly illustrates the functionalist perspective, which asserts that mental processes can be defined by their roles within a whole cognitive system.

Contemporary artificial intelligence is increasingly integrating functionalism via the application of artificial neural networks and deep learning algorithms. Despite being influenced by biological neurones, these networks do not reflect the brain's actual structure. They perform cognitive functions through layers of interconnected units that concurrently process inputs, modify internal states, and generate outputs. These units are tasked with executing these responsibilities concurrently. The relevance lies not in the biological analogies of these systems, but in their functional performance, which aligns with the concepts of functionalist psychology.

The concept of strong artificial intelligence posits that machines could possess mental states if they attain the requisite level of functional organization, a notion also endorsed by the theory of functionalism. This idea bolsters the notion of robust artificial intelligence. According to this perspective, an artificial intelligence system that can perceive, reason, learn, and act in a manner functionally analogous to human cognition can contain beliefs, desires, and potentially consciousness. This is due to the system's capacity to perceive, reason, learn, and act in a manner analogous to human cognition. Behaviourism, which confines mental processes to observable actions, and identity theory, which associates mental states exclusively with biological brains, are two contrasting approaches to this perspective. Both viewpoints endeavour to elucidate mental states by correlating them with physical brains.

Semantic networks, conceptual graphs, frames, rule-based systems, and procedural models exemplify the representational and computational techniques frequently employed by artificial intelligence (AI), alongside logical reasoning strategies. These approaches offer practical and diverse methods for encoding and manipulating knowledge, enabling machines to manage ambiguity, context, and the processing of vast quantities of information. In this context, artificial intelligence systems often surpass both the neural network framework and numerous Western epistemic traditions. These traditions tend to impose excessive importance on identifying the knower or the specific context in which knowledge is obtained. Although these issues are substantial from a philosophical perspective, they may hold less significance for advanced artificial intelligence systems, which can operate effectively without a cohesive, conscious subject of knowledge.

Western epistemology contributed substantially to the creation of numerous conceptual instruments that facilitated the advancement of artificial intelligence. The structure, reasoning, and assessment of belief have been focal points historically, originating with Greek logic and

extending into contemporary notions of justification. Artificial intelligence can be perceived as an extension of the pursuit to understand and replicate knowledge through formal methodologies. Conversely, persistent dialogues over meaning, consciousness, and uncertainty underscore that philosophical issues remain unresolved.

Knock your Brain

1 Which of the following questions is central to epistemology?

a) What is beauty?

b) What is justice?

c) What can we know and how do we know it?

d) What is the good life?

2 Which philosopher is most closely associated with empiricism?

a) René Descartes

b) Immanuel Kant

c) John Locke

d) G. W. F. Hegel

3 According to John Locke, the human mind at birth is like

a) A mirror

b) A blank slate (tabula rasa)

c) A rational structure

d) A divine spark

4 Identity Theory in philosophy of mind claims that

a) Mental states are patterns of behavior

b) Mental states are identical to brain states

c) Mental states are illusions

d) Mental states are non-physical substances

5 Identity Theory is also known as

a) Dualism

b) Functionalism

c) Mind–Brain Identity Theory

d) Logical Positivism

6 Which philosopher is most closely associated with the development of Identity Theory?

a) Gilbert Ryle

b) J. J. C. Smart

c) Hilary Putnam

d) Daniel Dennett

Unleash your potential

1 What is Identity Theory in the philosophy of mind?

2 Elucidate the theory of Behaviourism within the realm of the philosophy of mind.

3 Examine the principal characteristics of Behaviourism, accompanied by appropriate instances.

CHAPTER IV
Functionalism

It was in the middle of the twentieth century that the concept of functionalism evolved as an alternative to the mind–brain identity theory and behaviourism. Today, functionalism is widely regarded as one of the most influential views in the entire field of philosophy of mind. As opposed to characterising mental states based only on their physical makeup or on the actions of the observer alone, functionalists define mental states in terms of the roles that they play within a system. This is in contrast to the traditional practice of describing mental states. It is possible to get an understanding of a mental state by witnessing its behaviours, which includes how the condition is brought about by inputs, how it interacts with other mental states, and how it results in behavioural outputs. A great number of philosophers played a significant role in the development and refining of functionalism, which in turn affected contemporary debates in the domains of philosophy, cognitive science, and artificial intelligence as a result of their contributions.

Hilary Putnam who is considered to be one of the founding figures of functionalism. Putnam introduced the idea of machine-state functionalism in the 1960s. This proposed that the human mind is analogous to a computer program, and that the brain is analogous to hardware. Putnam maintains that mental states are comparable to computational states in the sense that, rather of being defined by their physical material, they are dependent on the functional interactions that exist between them. The theory of multiple realisability, which asserts that the same mental state can be realised in other physical systems, such as human brains, animal nervous systems, or even artificial devices, is the most influential contribution that he has made. This idea was developed by him. The identity hypothesis, which argued that certain mental states are similar to specific brain states, was open to a direct challenge as a result of this idea. As part of the process of establishing functionalism as a powerful and adaptable account of the mind, Putnam's work played a significant role.

Jerry A. Fodor is another renowned functionalist philosopher who is credited with bringing together the functionalist philosophy and a strong commitment to mental representation into a coherent philosophical framework. Fodor was the one who came up with the Language of Thought (LOT) theoretical framework. An internal symbolic language that is directed by syntactic rules is said to be the medium through which thinking takes place, according to this

hypothesis. According to Fodor, mental states are functional states that involve representations that hold semantic information. These representations carry out mental processes. His interpretation of functionalism, which placed a focus on the computational side of cognition, had a tremendous impact on the field of cognitive research. His interpretation of functionalism was particularly influential. In order to better understand the manner in which mental states are accountable for behaviour, Fodor argued that functional explanations are necessary. This, in turn, strengthened the capacity of functionalism to provide an explanation for behaviour.

It is generally agreed that David Lewis was the one who first developed the method of psychology that is today known as analytic functionalism. Regular psychology describes mental states in terms of the causal functions that they play, and this method describes mental states in terms of those functions. The position of ideas such as "belief," "desire," and "pain" within a network of relations that connect perception, action, and a variety of other mental processes is what Lewis considers to be the defining of these notions. Following the description of these responsibilities, it is feasible to determine the states that occupy these roles by comparing them to their appropriate physical states. By proving that mental notions can be expressed functionally without rejecting their physical manifestation, Lewis's approach was an attempt to bring functionalism and physicalism into harmony with one another. This was by demonstrating that the two schools of thought are compatible with one another.

Daniel Dennett offered a pragmatic and convincing defence of the functionalism philosophical school of thinking by contributing his concept of the intentional posture to the discussion. Dennett asserted that a suitable way for predicting the behaviour of a system is to assign the system's beliefs, goals, and intentions to the system itself. This holds true regardless of whether or not these states exist as private entities that are located within themselves. In accordance with this point of view, mental states are patterns that are able to be defined at the level of functionality. Because Dennett's work extends the framework of functionalism beyond the sphere of humans and into the realms of animals, machines, and artificial intelligence, it poses a challenge to the conventional concepts of consciousness and mentality that have been prevalent in the past.

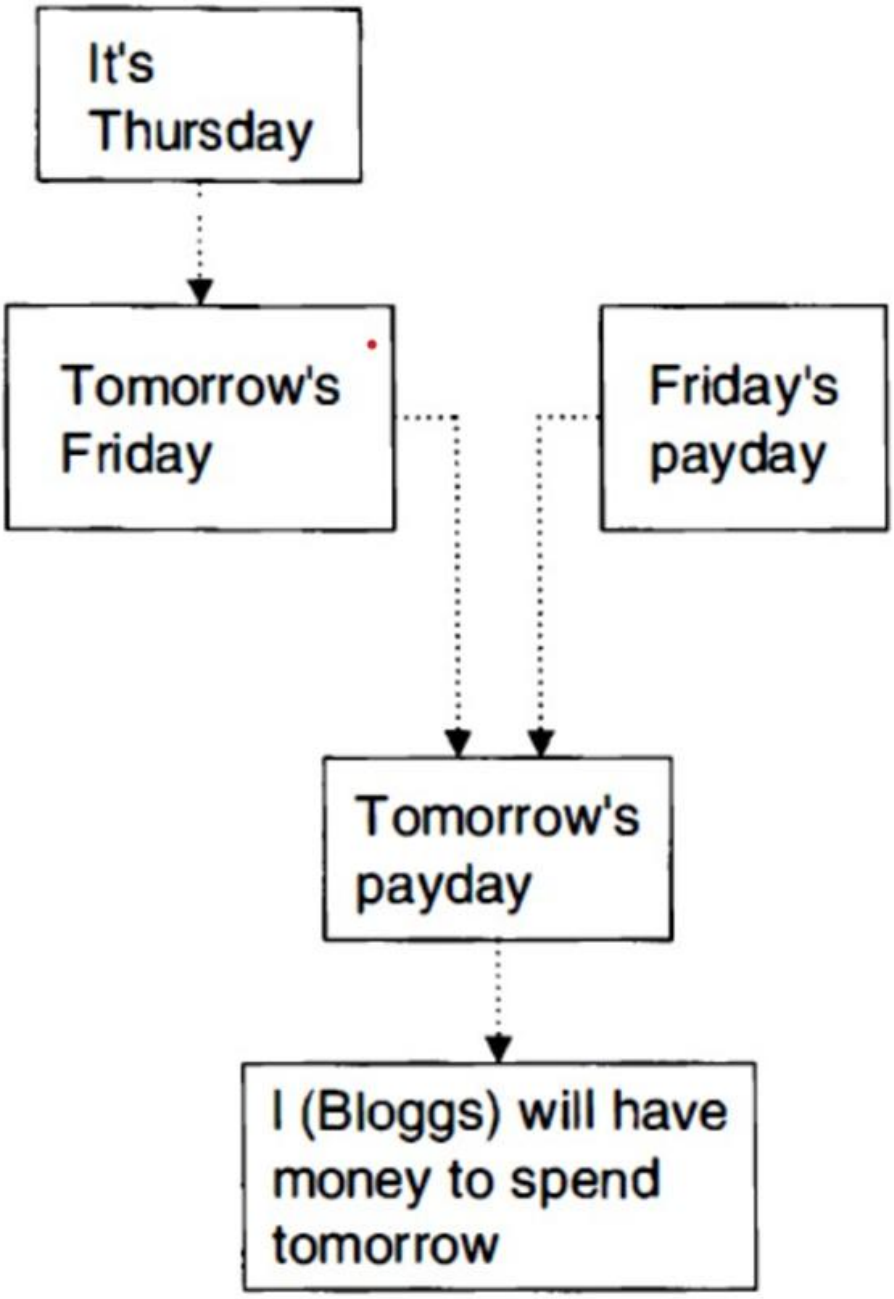
An additional notable contributor to this topic is Sydney Shoemaker, who, within the context of a functionalist framework, focused on mental causation and self-knowledge. On the other hand, Shoemaker contended that mental states must have actual causal capacities, and that functionalism may be able to account for this without reducing mental qualities to basic behavioural dispositions. In addition to this, he claimed that this is entirely consistent with the objective of functionalism. He was able to address criticisms that functionalism ignores

subjective experience as a result of his work, which contributed to the improvement of the logical foundations of the concept of functionalism. The functionalist philosophical perspective asserts that mental states are determined by the causal interactions that they have with one another, as well as by the sensory inputs and behavioural outputs that they experience. This is in addition to the fact that mental states are defined by individual experiences.

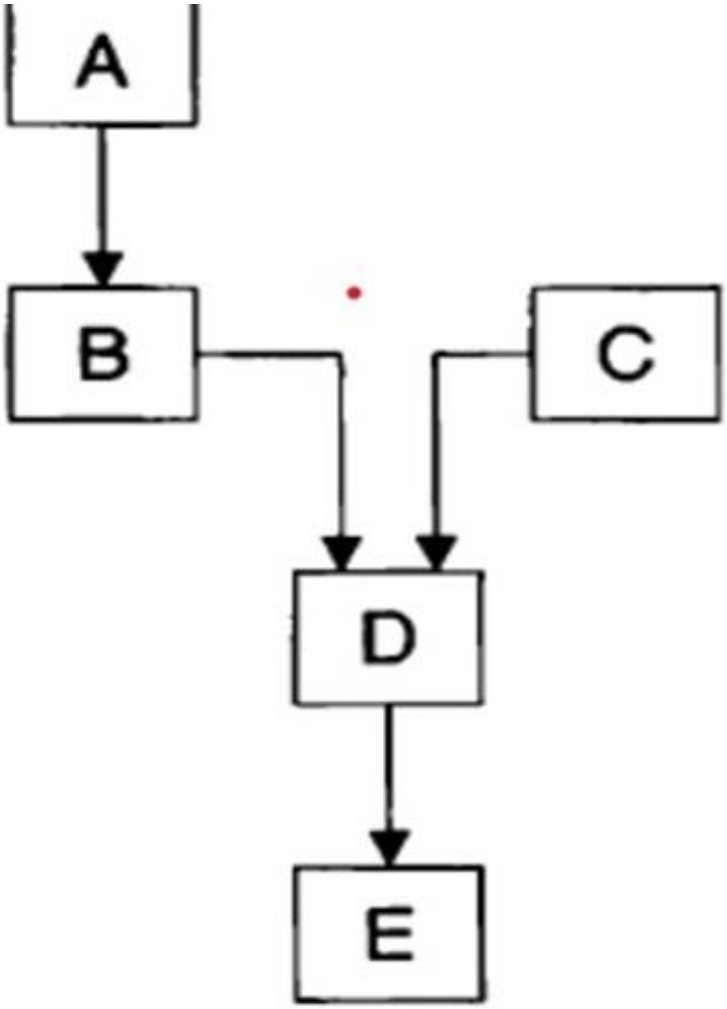
A significant theoretical accomplishment in the field of analytic philosophy during the entirety of the twentieth century is the functionalist school of thought. In addition to this, it is the conceptual foundation upon which a significant amount of research in the field of cognitive science is built. A functionalist approach to philosophy is based on the concept that many entities can be adequately defined not by their composition but by the functions that they perform. This is the theoretical foundation of the functionalist method. The capacity of the knife to cut is what differentiates it from other types of knives; the difference between steel and plastic is immaterial to the classification of the knife as a knife. This is true regardless of the material composition of the knife. A similar assumption is relevant to mental states and processes, according to the functionalist, who thinks that this is the case. It is not the material components of a given mental state, such as brain activity, that constitute its essence; rather, it is the functions that it involves and the kinds of causes and effects that it generates that establish its essence. In the theory of functionalism, it is proposed that mental states are distinguished by the functions that they normally perform, which are sometimes referred to as their "functional" roles. The viewpoint of a comprehensive functionalist is to assert that the causal function of a mental state (M) is the factor that determines all of the qualities of M. This is the position that all comprehensive functionalists support. In light of this, the causal function that M possesses is what determines the content of M. Using a functional role theory of content, one can gain an understanding of the process that is responsible for the establishment of causal roles. It is common practice to carry out the process of determining functionalist role theories of content in this manner. As a general rule, move forward. One have to take into consideration the fact that the mental states of an individual normally produce a network of causal relationships or connections. Take for instance the standard approach to the situation. For instance, Bloggs's conviction that it is Thursday led him to the conclusion that tomorrow is Friday; his belief that tomorrow is Friday, in conjunction with his belief that Friday is payday, resulted in his assumption that tomorrow is payday; and his belief that tomorrow is payday caused him to believe that he will have money to spend over the course of the following day. There is, in general, a causal network that contains mental states, and there is also an inferential

network that encompasses the substance of those beliefs. Both of these networks are examples of networks. At this point, we have merely acknowledged the existence of two networks: a causal network that connects mental states and an inferential network that associates the contents of mental states. Both of these networks are inferential networks. The substance of Bloggs' ideas in the example is presented as a cohesive argument, proceeding from a sequence of premises to the conclusion, which is that "I (Bloggs) will only need to spend tomorrow." It is crucial to observe that this line of reasoning is presented in a way that is consistent with the argument. It is the contention of those who subscribe to functional role theories of content that individuals attribute content to mental states by creating a correlation between the inferential network and the causal network.

An inferential network



A causal network involving five beliefs (A-E)



The scenario that is associated with the perception of pain is characterised by activation of the C-fiber in humans, whereas in squids, it is indicated by activation of the D-fiber. In humans, the C-fiber is activated when feeling pain. According to the functionalist hypothesis of the brain, it is believed that the activation of C-fibres in humans serves the same role as the activity of D-fibres in Flipper. According to this point of view, the presence of an internal state that

performs a specific purpose is what distinguishes the sensation of pain from other states of being. When an internal situation is the outcome of a bodily harm, it is said to be performing the "pain function." This function leads us to scream "ouch" and massage the area that is afflicted by the pain. It is the contention of the functionalist view that the experience of pain is a consequence of having an internal condition that is brought on by a physical harm. In response to this circumstance, we find ourselves uttering the phrase "ouch" and massaging the region that is being impacted. Putting this into a broader framework, the theory of functionalism posits that in order to have mental state M, one must also have an internal condition that satisfies the 'M-function.' This is a requirement for having mental state M.

Antibiotics are a class of chemical that eliminates bacteria that are harmful to the patient to the greatest extent feasible while causing the patient the least amount of harm possible. Penicillin is considered to be an antibiotic by medical specialists due to the fact that it removes hazardous microorganisms while inflicting the patient the least amount of harm possible. The antibiotic classification is given to erythromycin because of its capacity to eradicate pathogenic germs while inflicting the patient only a minimal amount of harm. These two antibiotics, penicillin and erythromycin, on the other hand, have chemical structures that are extremely unlike to one another..

When it comes to a number of significant aspects, the philosophical theory known as functionalism suggests that mental states are analogous to carburettors and antibiotics. The purpose of a carburettor is to carry out the "carburettor work," whereas the role of an antibiotic is to carry out the "antibiotic job." When compared to an antibiotic, a carburettor is defined by its function of carrying out. In a similar vein, the distinguishing characteristic of a certain mental state is the degree to which it is able to perform the functions that are associated with that state.

In the context of the concept of many realisations, it is an idea that suffering can present itself in a number of different ways. The following is a list of the numerous manifestations of pain, as viewed from the functionalist point of view. In the field of philosophy of mind, the term "functionalism" refers to a condition that takes on the role of suffering. In accordance with this idea, an organism is considered to be in a state of experiencing pain if it either possesses an internal state that is capable of performing the function of pain or if it is afflicted with a condition that takes on the function of pain.

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Inputs, outputs, and internal links are the distinctive qualities that are associated with the pain role. Instances such as walking on a tack, fracturing a leg, and boiling your hand are all instances of scenarios that have the potential to inflict significant pain. Inputs are the conditions that are responsible for causing discomfort. A person's outputs are the behaviours that are a direct result of experiencing pain. Some examples of outputs include, but are not limited to, shouting, rubbing the affected area, and vocalising "ouch." Pain and a range of mental states are represented by the internal connections, which are the causal interconnections between the two. The causal relationship that exists between pain and anxiety, for instance, states that acute pain often induces anxiety. This is one example of what they contain. Pain is one of the conditions that can be brought on by walking on a tack. Pain frequently brings about thoughts of dread and produces statements such as "ouch." These sensations can be brought on by walking on a tack.

Although functionalism accepts that it is impossible to directly correlate different varieties of mental states with different types of behavioural dispositions, it nonetheless strives to characterise mental states through behaviour, although in a roundabout fashion. According to the functionalist theory, the role that a mental state plays in the causal chain can be broken down into three distinct types of interactions that are deemed to be causal.

The way in which light that enters the eyes forms a visual impression is an example of how various characteristics of a subject's surroundings might first produce particular mental processes. This is because light is able to penetrate the eyes. Secondly, it is possible for certain mental states to have a causal interaction with other mental states that are present within the same individual. In order to illustrate this point, consider the fact that a feeling of discomfort in the leg may provide the impression that the limb in question has been harmed.

When it comes to the actions that an individual engages in with their body, certain mental states can have a causal influence on those activities. For instance, a person who feels that they have damaged their leg and subsequently has a great desire to alleviate the pain that they are experiencing is an example of this phenomenon. In the event that this occurs, the person may end up touching the injured area or removing their leg from a potentially dangerous situation.

According to functionalists, the categorisation of a mental state into a specific category is not fundamentally dependent on the mental state's underlying material makeup; rather, it is based on the mental state's functional role within the system to which it is attached. They believe that this is the most important factor in determining the classification of a mental state. The theory of functionalism is consistent with the idea that individuals and groups may exhibit sufficient

similarity in their functional organization to possess comparable mental states. This idea is consistent with the concept of functionalism. According to the functionalist theory, mental states are the bearers of a variety of causal functions. This is something that we have observed up to this point in our comprehension of the theory concerning mental states. Furthermore, we made the remark that, in principle, the roles that are inherent to various mental states can be completed by a wide variety of states. This was a significant finding for us. Because of this, functionalism is able to shed light on the infinite variety of mental states that are realisable. Now that we have arrived at this stage, we will study the link that exists between functionalism and the following theories: (i) type identity theory; (ii) restricted type identity theory; and (iii) token identity theory.

According to the functionalist hypothesis, the reason an organism is able to experience pain is because it possesses a state that is capable of carrying out the function of pain. It is a conceivable that the function of pain could be provided by a state of a material that is not physical at this point in time. This is a possibility that is entirely plausible. As a result of this, the statement that the functionalist substance dualism is accurate is one that can be considered plausible. They are of the opinion that it is highly conceivable that the actual physical states of the brain are responsible for carrying out the many causal functions that are associated with the various mental states. If functionalism is accurate, then there is a very good chance that a particular shape of the identity theory is authentic. Because of the close connection between the two ideas, this is the result.

Functionalism And Brain State

When attempting to characterise a state of the brain, it is possible to use the specific function of the brain. In the event that a person is experiencing pain, it is considered that they are going through an unpleasant brain state. Within a functionalist framework, it is possible to establish a wide variety of realisability, each of which can be considered distinct. It is possible, for instance, to characterise states of the brain such as hunger or pain by describing the functional duties that they have. Jaegwon Kim provides an instance of the notion of varied realisability in functional roles by using the mousetrap as an example. He is particularly well-known for his forceful criticism of non-reductive physicalism, which has made Jaegwon Kim one of the most influential contemporary philosophers of mind. Not only is the mouse trap analogy one of his numerous analogies, but it has also gained a lot of notoriety for its ability to convey the dilemma of multiple realisations and the pressure toward reduction. Kim is attempting to demonstrate, by means of this straightforward illustration, that higher-level characteristics might not possibly possess independent causal powers apart from their physical realisations.

It is a common argument among functionalists that mental states should be associated not with the components that make them up, but rather with the functions that they carry out. Pain, for example, is a mental state that may be defined by the manner in which it is caused, the manner in which it interacts with other states, and the manner in which it creates behaviour. Functionalists argue that mental states can be realised in a wide variety of materials, including human brains, animal brains, and even computers. This is due to the fact that different physical systems may execute the same function. This is the concept of having numerous aspects of reality.

In his statement, he does not refute the notion that several physical structures can carry out the same job. On the other hand, he raises the question of whether or not this demonstrates that higher-level qualities are irreducible or autonomous.

Additionally, Kim provides an explanation of the concept. In certain instances, a mousetrap will be equipped with a straightforward mechanism for activating the trap, but in other instances, it will be a wire cage that will close when sensors are activated. The incorporation of lasers or hydraulic power could be considered, depending on the specifics of the situation. Any device that is capable of capturing a mouse is regarded to be a mousetrap, regardless of its size, shape, or mechanism. This includes any and all devices that are capable of trapping mice. There is a connection between the states of the mind and the states of the brain, as the identity theory proposes as a possible explanation. There is a single instance of the category "C-fiber firing at the same time" that corresponds to each and every single case of the category "pain." The activation of the C-fibres and the experience of pain have a certain kind of identification with one another.

A concise explanation of the identity theory is that it claims that every type of mental state corresponds directly to a certain form of neurological condition. This is a statement that represents the theory. On the other hand, the identity theory does not require that every brain condition connects to a specific mental state in order to be valid. To be more specific, the states that are being taken into consideration are the mental states that the brain is currently experiencing. Furthermore, the identities should not be determined by only observing the individuals or by evaluating the meanings of the terminology that are pertinent to the situation. The brain state that is often created by the visual and aural cues of rain is precisely the same as the one that Thomas is convinced is occurring, which is that it is raining. A common occurrence is the coexistence of this brain state with other brain states that, among other things, result in the behaviour of retrieving an umbrella. Lewis posited that mental states are analogous to the

physical states that are present within us, and that these mental states serve particular functions that allow them to be considered causal. The upshot of this discovery is that the association between specific mental moods and specific physical conditions may vary from person to person within the same group. Also, this correlation may be different for different people. According to Lewis, the physical condition that corresponds to a specific mental state is dependent upon the causal roles that the state normally plays in the organism that possesses it. This is the point that Lewis is making. Pain is a unique form of brain state that is present in humans. All humans are capable of experiencing pain.

Functionalism And Six Features of Mental State

1There are particular states of the world that are the underlying cause of particular mental states. We have shown that the functional functions that are linked with various mental states are likely to be occupied by states of the brain. This is something that we have established. Therefore, the argument that certain mental states are caused by states of the world is only valid, according to the functionalist, if certain brain states are caused by states of the world

. 2. Various mental states are responsible for a variety of behaviours. Again, it is essential to bear in mind that it is quite possible that the states of the brain are the ones that occupy the functional functions that are specifically associated with mental states. This is something that should be kept in mind at all times. The argument that particular mental states are accountable for actions is only legitimate, according to the functionalist, if certain brain states are responsible for those actions. This is because the functionalist believes that action is a function of the brain.

3. It is possible to trace certain mental states back to the origin of other mental states. Given that mental states are states of the brain, the argument that certain mental states are responsible for other mental states is equal to asserting that certain brain states are responsible for other brain states. This is because mental states are states of the brain. Moreover, this is without a doubt the situation. Not only is it true that certain mental states are the cause of other mental states, but it is also true that the causal linkages between mental states can sometimes mirror the rational relations that exist between them. This is something that is true.

4. For certain mental states, it is feasible to be aware of them. Since the beginning of time, consciousness has always been a tremendous obstacle to overcome. It would appear that it is conceivable to think of a robot that possesses a central computer that is capable of occupying the functional role that is characteristic of pain, but does not experience it. This would be a remarkable achievement. It would appear that the functionalist approach faces a big obstacle in the form of consciousness if what you have stated is true.

5. There are things or events in the world that are related with certain mental states that people experience. From a theoretical standpoint, there are two distinct points of view regarding this topic. One of these, known as functional role semantics, is one of those that is quite at ease with the concept of functionalism. Alternatively, the other approach, which incorporates the causal theory of content, calls for additions to the underlying functionalist framework and, at the very least, some modifications. This is because the causal theory of content considers content to be a causal phenomenon.

6There is a continuous association between many different types of brain states and a wide variety of mental states, as researchers have discovered. It has been brought to our attention that it is not inconceivable that the functional functions that are distinctive of the various mental states be inhabited by brain states. Such a possibility has been brought to our attention. The functionalist viewpoint is compatible with the argument that there are systematic links between mental states and particular brain states. This is because the argument is based on the idea that there exist such ties.

Knock your Brain

1. Functionalism defines mental states mainly in terms of their:

A. Chemical composition

B. Physical location

C. Causal roles

D. Spiritual nature

2. Functionalism emerged largely as an alternative to:

A. Dualism only

B. Identity theory and behaviourism

C. Pragmatism

D. Existentialism

3. According to functionalism, a mental state is identified by:

A. Brain tissue

B. Input–output relations

C. Personal feelings only

D. Moral value

4. The idea that the same mental state can exist in different physical systems is called:

A. Supervenience

B. Eliminativism

C. Multiple realization

D. Parallelism

5. Which philosopher is strongly associated with functionalism?

A. B. F. Skinner

B. Hilary Putnam

C. René Descartes

D. David Hume

Unleash your potential

1.Elucidate the notion of functionalism within the philosophy of mind.

2.Examine the correlation between functionalism and artificial intelligence.

3.Elucidate the distinctions between functionalism, identity theory, and behaviourism.

CHAPTER V
Epistemological implications in Artificial Intelligence

Indian

In the era of computers of the fifth generation, there has been a significant amount of work devoted into the simulation of intelligent behaviour in machines. By using grammar-based computing, in which the Knowledge of Language (KOL) is sought to be given to machines for Natural Language Understanding (NLU) or Processing (NLP), and by using knowledge-based computing, in which the background/contextual or Real World Knowledge (RWK) is sought to be represented in machine for understanding real world situations and problem solving, the researchers are attempting to find a solution to this problem. Both of these approaches are being considered by the researchers. An increasing number of people are coming to the realisation that RWK representation is necessary for the simulation of consciousness in machines. With that being said, there has not been a single framework that has been able to be developed that is applicable to all sorts of knowledge, identities of the locus, object, and instrument of cognition. Many academics in the field of artificial intelligence are beginning to question whether or not the following statements are true:

* Knowledge can be represented
* Computers can encode and decode knowledge in everyday language
* Computers are capable of thinking and understanding
* Computers require knowledge in order to communicate with people or other machines

Although Panini's grammar is certainly applicable to RWK, it is actually an account of KOL that can be used as a model for other languages as well. This is because there are indisputable descriptions of both the KOL and the RWK in India, particularly in the Sanskrit language. It is possible to find one of the first accounts of RWK in the NN system, which is not too dissimilar to the accounts that the AL philosophers have provided. As a result of this, it is of the utmost importance to investigate what the NN system has to offer. It is to compare and contrast the Indian epistemological tradition, specifically the NN, with the one that is associated with artificial intelligence.

The western perspective on knowledge has traditionally centred on the question of whether knowledge is learned (posteriori) or given (a priori), which means "from what is before" instead of "from what is after." In contrast to the latter, which developed into the empiricist way of inquiry, the former ultimately evolved into the rationalistic mode of inquiry. Plato contends that there is a consciousness of absolute, universal concepts or forms that exist apart from any

individual effort to comprehend them. He says this awareness exists independently of any subject. In his approach to the acquisition of knowledge, Aristotle places a greater focus on logical and empirical approaches. Moreover, he acknowledges the viewpoint that knowledge is an understanding of fundamental truths that are universally applicable. All of the early rationalists, including Descartes, Spinoza, and Leibniz, believed that the most important source of knowledge was deductive reasoning that was founded on axioms. The early empiricists, such as Francis Bacon and John Locke, considered sense perception to be the primary source of information and the ultimate test of knowledge. Despite the fact that Berkeley shared Locke's views on the relationship between knowledge and ideas, he was opposed to the idea that ideas should be separated from objects. As an empiricist, Hume was not in agreement with Berkeley's conclusion that knowledge may be derived solely from philosophical concepts. Hume classified all knowledge into two categories: relations of concepts and the knowledge that can be found in mathematics and logic. This knowledge is precise and certain, but it does not convey any information of any type about the world. the knowledge that is obtained from the senses and the facts that are known. The debate between empiricists, rationalists, pragmatists, and constructivists persisted both throughout the Renaissance and after it had ended. The reflection-correspondence hypothesis, which is part of the empiricist school of thought, asserts that knowledge is the product of a reflection of external things onto our brain or mind throughout the learning process. It is their belief that the meaning is what the words signify, and that language, much like a mirror, reflects the things that are going on around us. If there is a correspondence between the relations between the representations of things and the relations between the things themselves, then the assertion stated is true. An indication of untruth is a reflection that is incorrect and distorted, which is a picture that does not correlate to any real thing in the world. Kant was the one who sought to develop a synthesis of the competing arguments that were presented by rationalists and empiricists. Knowledge is said to be the consequence of the organising of perceptual facts on the basis of inborn cognitive structures or categories, as stated by Kant. Space, time, things, and causality are all items that fall under the Kantian category. According to Distantie (2003), Kant differentiated between three distinct categories of knowledge.

Kant, Spencer, Bergson, Alexander, and Dawes Hicks are some of the philosophers who are credited with being the pioneers of the Act Theory of Knowledge. This theory attempts to explain knowledge in terms of activity. The behaviourists, on the other hand, refer to it as an activity of the body; nevertheless, contemporary European philosophers such as Moore and Broad have not accepted this idea. The Self-Subsistency Theory, which asserts that knowledge

is something that exists on its own, is supposed to have been proposed by Berkeley, who is considered to be the most prominent western philosopher (Bijalwan 1987).

There are two major fields of research in artificial intelligence: knowledge representation (KR) and information retrieval (IR). Knowledge can be characterised as objects and attributes, or even as an algorithm that contains objects and procedures, when approached from the perspective of artificial intelligence (AI). An interaction between "data" and "processes" or "abilities" and "memory storage" is stated to be the cause of the so-called "intelligent behaviour" that is observed. According to Jean-Raymond Abrial (1974), a database is a representation of the physical world as it is always changing. During a particular moment, the state of this model is a representation of the knowledge that it has obtained from the outside world. But knowledge is more than static encoding of facts in interacting with the world. The idea that knowledge of anything is the capacity to construct a mental model that accurately depicts the thing in question, as well as the actions that may be carried out by it and on it, is one of the fundamental premises of artificial intelligence (AI). Following this, individuals (or robots) are able to make predictions about what is likely to occur in the real world by testing actions on the model (Sowa in 1984). An earlier challenge faced by those working on artificial intelligence was determining whether to represent knowledge by using a declarative approach or a procedural method. For the purpose of knowledge representation, it has been finally decided that the most appropriate strategy would be one that combines a declarative approach with a procedural approach. Predicate calculus, semantic nets, frames, conceptual graphs, scripts, semantic primitives, and neural nets are some of the other well-known techniques for KR. These techniques follow a wide variety of strategies for KR. In spite of this, reducing the unknown phenomenon known as knowledge to a single or two definitions would not be an aim in and of itself. According to Hopcroft and Lipman (1979), not only would it be necessary to comprehend the entirety of the human cognitive process, but it would also be necessary to develop efficient methods for representing information.

pratyaksa (perception)

* anumana (inference)

* upamana (analogy)

* sabda (verbal testimony) and latter is sub-classified as

* samsaya (doubt)

* viparyaya (error)

* tarka (hypothetical argument)

* smrti (memory)

5.1. **Comparison with AI**

The following sub-sections define each of the above sub-classifications of invalid and valid

knowledge and give their AI parallels--

5.1.1. apramana--the invalid knowledge

It is defined as false knowledge, doubtful or illusionary. The four types of invalid knowledge are

defined and compared with AI as follows:

5.1.1.1. smrti (memory)

* is impressions left by previous knowledge

* it is non valid because the object of knowledge is absent at the time of remembrance

* AI explanation

* memory as gaps in knowledge or probabilistic knowledge which must be corrected before

processing starts and conclusion is arrived at

* such gaps or incomplete knowledge may be replaced by new knowledge [right arrow] nonmonotonic knowledge (McDermott & Doyle 1980)

* static memory [right arrow] knowledge of object (WHAT)

* short term : memory in current focus

* long term: memory for future use

* procedural memory [right arrow] knowledge of behavior (HOW)

* short term

* long term

5.1.1.2. samsaya (d * conflicting judgment on the precise character of an object of cognition (Bijalwan 1987)

* neither true or false, only wavering judgment

* AI explanation

* ambiguity resolution and bug handling come close to samsya

5.1.1.3. tarka (hypothetical judgment)

* AI explanation

* Heuristics

* when reality is not known, truth is ascertained trying alternatives

* atmasraya (petito principi)

* if A [right arrow] A then A must be different from A

* anyonyasraya (dependent tarka)

* A[right arrow]B[right arrow]A

* cakraka (circular logic)

* recursion in AI

* A[right arrow]B[right arrow]C[right arrow]A

* anavastha (regressus ad infinitum or transitivity)

* A[right arrow]B[right arrow]C[right arrow]D ...

* tadanya-budhiat-prasanga (reduction ad absurdum)

* proves a proposition by negating the contradiction of the proposition

* A[right arrow]-(-A) or + [right arrow] -(-)oubt)

5.1.1.4. viparyaya (error)

* manifestation of a real object in another * positive misconception, incomplete knowledge, gaps in knowledge * AI explanation * can be compared to bugs and inconsistencies

5.1.2. pramana--the valid knowledge

The following section discusses the means to valid knowledge and compares them with AI

5.1.2.1. **pratyaksa (perception)**

* Five senses--sight, smell, taste, hearing, touch (mind [right arrow] sixth sense for cognition of

feelings like pleasure, pain etc)

* six ways of perception for six kinds of objects

* substance, quality, universal quality, sound, sound-ness, nonexistence

* AI [right arrow] icon, schema and percept

* icon [right arrow] temporary record of a sensory input that the brain keeps

* schema [right arrow] blue print of the mental model

* percept [right arrow] assembling units which are to be assembled according to the schema to

create a mental model

* Sowa (1984) calls the search mechanism "associative comparator" which must have the following

characteristics:

* associative retrieval

* brain is like a computer, retrieves data by an address in the storage by matching the best pattern

* top-down match

* perception searches for percepts that match the overall pattern of an icon

* stimulus constancy

* stimuli from same external object are recognized as equivalent despite varying size, brightness etc

* distributed storage

* images are not stored at a specific point, but distributed

5**.1.2.2. anumana (inference)**

* Gautama presupposes perception

* Jayanta gives five processes of inferring

* Perception of reason

* Remembrance of the universal concomitance

* Judgment that the subject of inference contains sense which, is concomitant with the object

* Knowledge of the consequence, and

* Judgment that the consequence is worthy of being accepted or rejected

* AI

* inference is very important and strong point of predicate calculus

* axioms [right arrow] theorems

* new facts can be deduced from axioms using rules of inference similar to ones given by Gautama

Gautama gives 3 ways to infer:

* psrvavat (deduction)

* cause [right arrow] effect.

* cloud [right arrow] rain

* "Gauri eats grass" from "Gauri is a cow" and "all cows eat grass" --

(inst gauri cow)

(forall X (if (inst X cow) (food X grass)))

(food gauri grass)

* sesavat (abduction)

* process [right arrow] explanations

* effect [right arrow] cause

* river is swollen [right arrow] there could have been rain

* Abduction has the following paradigms

From: b

(if a b)

Infer: a

* unlike deduction, abduction is not a legal interference. It can lead to false conclusions--

from: (feels nervous Ira)

(forall (X) (if (is sick X)

(feels nervous X)))

infer: (is sick Ira)

* samanyatod (induction)

* infers consequent from the antecedent, which is neither cause nor effect

* while induction can take several forms, the most common is

from: (P a), (P b),...

infer: (forall (X) (P X))

* although this is not a sound inference, induction like abduction, is very useful in everyday life where

it is more commonly known as learning

* for example, if we see a lot of leaves of green color, we might infer that all leaves are of green color

from: (if(inst leaf-1 leaf) (colour leaf_1 green))

(if(inst leaf-2 leaf) (colour leaf_2 green))

Infer: (forall (X) (if(inst X leaf) (colour X green)))

* Vacaspati divides inference into vita and avita (Bijalwan 1987). The former is based on universal

agreement in presence. As an example, we can infer

from: whatever has smoke has fire the hill has smoke

infer: the hill has fire

* avita is based on universal agreement in absence

from: whatever is non-different from other objects has no

smell. Earth has smell

infer: Earth is different from others.

* Prasastapada has svarthanumana and pararthanumana as two types of inference.

* Svarthanumana

* the premises are known from our own experience

*pararthanumana

* They are discovered by one man and imparted into another through language.

* Therefore, there is greater chance of error in pararthanumana which is based on words.svarthanumana is a mental process which is divided into data and samanyatodata by Prasastapada(Bijalwan 1987).

* Udyotakara gives kevalanvayin (K), kevalavyatirekin (KV), and amayavyatirekin (AV) as three types of inference. If a middle term is only positively related to the major term it is called K. Here. Thereason exists in the subject and similar instances, and is devoid of dissimilar instances. For example--we can infer "pot is nameable" from "pot is knowable" and "all knowable things are nameable"--

(property X nameable)))

(if (inst pot knowable)

(property pot nameable))

(property pot nameable)

In the case of KV, the middle term is negatively related to the major term. Reason exists in the

subject, but not in dissimilar instances. For example--

from: (no (not P) is (M))

S is M

infer: S is P

When the middle is positively as well as negatively related to the major term, and reason exists in the entire subject, and in all similar instances but does not exist in dissimilar instances, the case is that of AV, as "sound is non-eternal because it is produced like a jar". The inference of Fire from smoke is also of this kind. Jayanta does not however, accept this division of inference in to K, KVand AV by Udyotakara (Bijalwan 1987).

5.1.2.3. upamana (analogy)

* upamana is the means by which we gain the knowledge of a previously unknown object on thebasis of its similarity to another object previously well known

* Mimansists and the Vedantists also accept analogy as an independent source of knowledge

* AI

* analogy is too slippery to be seen as just another road to induction

* often, analogy matches are so slipshod that they merely suggest solutions to the problem ofknowledge acquisition in general, and for inductive learning in particular

* analogy is useful because it serves as initial framework for developing the new concepts

5.1.2.4. sabda (verbal testimony)

* Leaving aside the Carvakas, Vaisesikas and the Buddhists, all the other systems of Indian philosophy accept sabda as a distinct source of knowledge, but they also differ with one another with respectto its nature forms and a number of other aspects.

* Jayanta (Nyayamanjari)

* contains a long discussion on this problem

* justices the acceptance of sabda as a distinct means of knowledge

* gives serious thought to the philosophy of language

* tries to explain the views of his predecessors and evaluates

* the arguments of prominent scholars

* his account of sabda is as much relevant to grammar, rhetoric, and linguistics as it is to logic

* convincingly proves that words do not exist before their production

* also refutes the sphota theory of grammarians and argues that the relationships between words and meaning is conventional

* AI

* sabda can be compared to the concept of formal string and language.

* language is a set of strings made from some alphabet

if a fixed alphabet = E, then the language formed by

E is called [E.sup.*]

ifE= {a} then

[E.sup.*] = {e. a. aa. aaa. ...}

Both the Naya system and the AI system are subject to comparison and contrast. As has been demonstrated, the Naya system shares many similarities with the AI in terms of how it handles valid and invalid kinds of knowledge. The AI, on the other hand, employs a variety of different methods in addition to logic and inference. These methods, including as semantic networks, conceptual graphs, frames, rules-based systems, and processes, offer suggestions for practical ways to represent information. This is in contrast to the NN and even western schools, which place an excessive amount of focus on fixing the identity of the locus of knowledge acquisition. The latter is perhaps not as important for artificial intelligence at the moment. In spite of this, there is a pressing need to conduct additional research into a number of Indian systems, such

as the neural network (NN) system and others, such as Buddhist logic, in order to develop appropriate models of human cognition that can be utilised by artificial intelligence.
The way in which Nyāya differentiates between valid and invalid information is similar to the way in which artificial intelligence differentiates between accurate forecasts and errors. False inference, also known as mithyājñāna, may be compared to algorithmic bias or misclassification in artificial intelligence (AI). In the Nyāya approach, errors are corrected by reasoning, which is similar to the training of artificial intelligence models, where recurrent learning helps reduce errors. For the purpose of verifying truth claims, artificial intelligence could utilise logical validation in the style of Nyāya.I.e. New artificial intelligence mistake-detection frameworks can be inspired by Nyaya's classification of perceptual and inferential errors, which is used in error analysis. There are similarities between the hierarchical knowledge graphs in artificial intelligence and the structured approach to cognition that Nyāya takes. Co-Reasoning between Humans and Artificial Intelligence: The approach of combining human logical insight (Nyāya) with machine inference (AI) has the potential to improve interpretability and accountability. The logical structure of Nyāya places a strong focus on the cause-effect link, also known as vyāpti, which states that a particular cause consistently results in a particular effect. For instance, smoke suggests fire. When it comes to artificial intelligence (AI), this is comparable to the concept of causal reasoning and explainable AI (XAI), which involves ensuring that the output of an AI is not only statistically accurate but also logically and causally warranted. This idea is at the core of Nyāya logic.

When comparing the Nyāya–Nyāya (NN) system with the artificial intelligence (AI) systems of today, it is important to note that each system approaches knowledge differently. It has been demonstrated that the neural network (NN) system has a number of significant similarities with artificial intelligence (AI), particularly in its systematic differentiation between legitimate (pramā) and invalid (aprāmā) kinds of knowledge. The two frameworks acknowledge that knowledge is not simply the accumulation of information; rather, it must fulfil certain characteristics of trustworthiness, correctness, and justification in order to be considered knowledge as such. On the other hand, whereas the NN system is primarily dependent on organised logical inference and properly stated epistemic categories, contemporary AI systems make use of a wider variety of approaches that go beyond the traditional methods of logic and inference alone.

With that being said, the significance of Indian epistemological frameworks for artificial

intelligence research is not diminished in any way. To the contrary, there is still a significant requirement to engage with Indian philosophical traditions, such as the NN system and Buddhist logic, in a more in-depth manner in order to develop models of cognition that are more complex and subtle. For example, Buddhist theories of inference, momentariness, and dependent origination provide alternate methods of explaining reasoning processes. These theories do not presuppose the existence of a permanent self, which is consistent with the non-subjective character of artificial intelligence systems. Likewise, the rigorous examination of mistake, doubt, and verification that the NN system does can provide valuable insights for the development of artificial intelligence systems that are more resilient, transparent, and epistemically accountable.

WEST

The status of machine knowledge is a significant epistemological conclusion that arises from artificial intelligence. Understanding knowledge as a human effort that involves belief, truth, and justification has been the conventional understanding of the concept. It is possible for artificial intelligence systems to store knowledge, draw conclusions, learn from data, and provide predictions. In light of this, the question arises as to whether artificial intelligence systems actually know things or only manipulate symbols and figures. People who adhere to a strict epistemological perspective contend that artificial intelligence (AI) does not contain understanding and consciousness, and as a result, it does not possess knowledge in the full human sense, but rather only derived or instrumental knowledge.

Moreover, artificial intelligence reshapes disputes concerning the origins of knowledge. Knowledge that is a priori, which is independent of experience, and knowledge that is a posteriori, which is gained from experience, are two types of knowledge that are distinguished by classical epistemology. Models of knowledge acquisition that are empiricist in nature are similar to machine learning systems since they rely largely on data and experience. A priori, on the other hand, refers to the nature of the algorithms, logical frameworks, and mathematical models that guide artificial intelligence. Artificial intelligence (AI) blurs the traditional epistemic boundaries by combining aspects of rationalist and empiricist approaches.

Another crucial implication is to the trustworthiness and explanation of the statement. Reasons, evidence, and critical reflection are the means by which human epistemology seeks to justify views. Particularly when it comes to deep learning models, artificial intelligence systems frequently generate outputs that are extremely accurate yet difficult to explain. Epistemic opacity is a problem that arises as a result of this, in which even developers are unable to completely account for how an artificial intelligence arrived at a conclusion. As a consequence of this, concerns arise regarding the epistemological justification of beliefs or recommendations that are formed by means of artificial intelligence.

Additionally, artificial intelligence brings about new kinds of epistemic authority and trust. AI systems are having an increasingly significant impact on decision-making in a variety of sectors, including medical, law, economics, and research. This causes the epistemic responsibility to move from the individual human knowers to the complex socio-technical systems around them. It is now necessary for epistemologists to enquire about who is accountable for the knowledge claims that are generated by artificial intelligence: the system, the programmer, the data, or the user, and how trust ought to be allocated.

The development of artificial intelligence has an additional impact on the boundaries of knowledge. Artificial intelligence has the ability to comprehend enormous datasets that are beyond the cognitive capabilities of people, which enables new discoveries and patterns that humans could never see. Artificial intelligence is, however, constrained by factors such as the quality of the data, biases, and the architecture of the algorithm. These limits serve as a reminder that the information behind artificial intelligence is neither objective or flawless; rather, it is formed by human values and assumptions that are entrenched in its development.

When it comes to social and collective epistemology, artificial intelligence has ramifications. Collaborations between humans and machines are making an increasing contribution to the production of knowledge. The significance of ideas like distributed cognition, extended mind, and common knowledge is beginning to regain its previous prominence. The incorporation of artificial intelligence into the epistemic environment brings about a transformation in the manner in which societies generate, validate, and distribute knowledge.

The fact that identity theory is dubious of powerful artificial intelligence is a significant implication that it has for the field of artificial intelligence. If some mental states are the same as certain biological brain states, then it is impossible for computers made of silicon, circuitry, and code to actually possess mental states. This is because computers are not capable of developing their own mental states. The simulation of intelligence or awareness by artificial

intelligence systems is a possibility; nevertheless, from this perspective, it is important to note that these systems do not genuinely think or feel. The conclusion that might be drawn from this is that the notion that artificial intelligence can ever have authentic feelings, experiences, or beliefs is called into question.

Furthermore, identity theory places a considerable amount of emphasis on the biological underpinnings of cognitive processes. This argues that in order to acquire true intelligence through the field of artificial intelligence research, it would be required to design systems that could closely duplicate the physical structure and activities of the human brain. This would be necessary in order to reach genuine intelligence. Models of computing that are simple or algorithms that are abstract would not be sufficient in this situation. Furthermore, as a result of this, identity theory offers philosophical support to a number of approaches, such as artificial intelligence that is inspired by neurological processes, brain emulation, and biologically realistic neural modelling.

the limitation that it imposes on the concept of multiple realisability through its implementation. In contrast to functionalism, which allows mental states to be realised in a broad range of other physical systems, identity theory restricts mental states to particular subtypes of physical states, notably those that are associated with the human brain. The upshot of this is that identity theorists may argue that artificial intelligence does not possess true consciousness since it does not have the same physical identity as a human brain, even if it behaves exactly like a person at all times. This is because artificial intelligence does not share the same physical properties as a human brain.

On the other hand, identity theory does not in any way undermine the practical applications of artificial intelligence in the real field. From this perspective, artificial intelligence (AI) can still be of immense use as a tool for replicating human cognition, anticipating behaviour, and offering support with decision-making. These are all areas in which AI can be extremely beneficial. Artificial intelligence models have the potential to supply neuroscientists with simulations that can be tested, which can be of great assistance to them in comprehending how the operations of the brain function. Despite the fact that the models themselves are not conscious, this is nevertheless the case.

Because it argued that intelligence can be linked with behaviour rather than with the mental processes that occur within an individual, behaviourism had a significant impact on the early concept of artificial intelligence (AI). This was due to the fact that behaviourism proposed that intelligence can be linked with behaviour. The behaviourist school of thinking holds that an

artificial system can be considered intelligent if it exhibits behaviours that are indistinguishable from those of a human being. This is because the behaviour of an artificial system is indistinguishable from that of a human being. Alan Turing's Turing Test is a test that examines the intelligence of computers solely on the basis of observable language behaviour, without making any reference to the mental states of the machines themselves. There is a significant connection between this notion and the Turing Test, which is a test that analyses the intelligence of machines. Therefore, behaviourism provided a conceptual justification for evaluating artificial intelligence only on the basis of its performance, rather than on the basis of its fundamental architecture.

In the field of artificial intelligence, behaviourism is characterised by the fact that it focuses a strong emphasis on the connections between input and output. In the early stages of the development of artificial intelligence systems and machine learning models, the stimulus–response framework that is a part of behaviourist psychology was taken into consideration. It was the intention of these systems to provide appropriate outputs in response to certain inputs. In particular, reinforcement learning is a form of learning that is reflective of behaviourist notions. This type of learning is characterised by the use of both rewards and penalties in order to shape the behaviour of an agent. In this sense, behaviourism played a significant role in setting the groundwork for the development of realistic artificial intelligence systems that are based on learning from its surroundings through feedback.

Behaviourism also conveys the suggestion that the existence of internal mental states is not necessary for intelligence to exist. For an artificial intelligence system to be considered intelligent, it is not required for it to have consciousness, comprehension, or subjective experience; rather, it is sufficient for it to behave in a manner that is consistent with intelligent behaviour. Weak artificial intelligence is a type of artificial intelligence in which machines simulate intelligent conduct without claiming true comprehension or intellect. This provides support to the establishment of weak AI, which is a sort of AI. As a consequence of this, behaviourism avoids engaging in complex metaphysical debates concerning the presence of mind and consciousness in machines.

The behaviourist approach to artificial intelligence, on the other hand, has consistently been the target of severe criticism. The argument that engaging in intelligent activity does not necessarily indicate true comprehension is one of the most major concerns that comes up. John Searle's argument in the Chinese Room poses a challenge to the behaviourist theory since it demonstrates that a system can create correct responses without fully knowing the meaning of those responses. Some individuals, on the other hand, are of the opinion that behaviourism does

not take into consideration internal cognitive processes like as representation, reasoning, and intentionality. These processes are essential for offering an explanation for higher-level intelligence.

Regarding the ethics of artificial intelligence, there is also a substantial impact. Artificial intelligence systems are unable to have internal experiences such as joy or anguish, and as a result, they are unable to sense moral damage. This is based on the assumption that behaviourism is accurate. As a result of this, artificial intelligence systems do not have any moral position, and they are completely responsible for ethical behaviour. The responsibility for ethical behaviour resides with humans. Despite the fact that this makes it simpler to make ethical decisions, it may become troublesome in the future when artificial intelligence systems behave in a manner that is more like to that of human beings.

The conclusion is that behaviourism has been a significant influence on the creation of artificial intelligence as well as the appraisal of its effectiveness. Because behaviourism lays an emphasis on visible behaviour, performance, and learning through reinforcement, this is the reason why this is the case. It was a valuable and significant paradigm for early research on artificial intelligence; nevertheless, the fact that it did not take into consideration the mental processes that occur within the mind inhibits its capacity to describe genuine knowledge and consciousness. This is despite the fact that it was a helpful and important paradigm. Although more complex cognitive and functional theories have taken their place, behaviourism continues to be an important historical and philosophical grounding for artificial intelligence (AI) as it continues to grow. This is the case despite the fact that these theories have taken their place.

In the field of Artificial Intelligence (AI), functionalism has been an important and influential feature in both the research and application of AI. Functionalism provides a significant conceptual foundation for artificial intelligence (AI) by advocating for the idea of multiple realisability. It is feasible to achieve the same mental state in a variety of different physical systems by utilising this method. Some examples of these physical systems include the human brain, the neurological systems of animals, and even artificial robots. As a result of this, researchers working in the field of artificial intelligence do not have to imitate the biological structure of the human brain in order to produce intelligent systems through their work. These individuals, on the other hand, focus their attention on replicating the functional organization of cognition, which is in complete accordance with the functionalist philosophical perspective. One of the oldest applications of functionalism in the field of artificial intelligence is symbolic or classical artificial intelligence, which is an example. The early models of artificial

intelligence were based on machine-state functionalism, which served as their primary origin of inspiration. According to these conceptions, the mind is a computer system, and the mental operations that occur are considered to be computational states. Artificial intelligence programs made use of symbols, rules, and algorithms in order to communicate their beliefs, goals, and intents to the public. In addition to these symbolic representations, functional activities such as thinking, planning, and problem-solving were developed during the process of development. This was done in order to reflect the functionalist view that mental states are determined by the causal functions that they perform within a system. This was done in order to indicate their significance.

Additionally, the rise of cognitive architectures such as SOAR (State, Operator, and Result) and ACT-R and (Adaptive Control of Thought—Rational) is evidence that functionalism is actively present. These systems seek to imitate human cognition by disassembling the mind into its component elements, which are functionally distinct from one another. These components include perception, memory, learning, and action, among others. What characterises a component is not the manner in which it is physically manifested; rather, it is the function that it performs that is the defining characteristic of the component. In a similar manner, this reflects the functionalist position, which maintains that mental processes may be characterised by the roles that they play within an overall cognitive system.

Artificial intelligence in the modern era is progressively incorporating functionalism through the utilisation of artificial neural networks and deep learning systems. In spite of the fact that they are subject to a significant amount of impact from biological neurones, these networks do not correspond to the physical structure of the brain. They instead carry out cognitive activities by means of layers of interconnected units that are responsible for simultaneously processing inputs, transforming internal states, and producing outputs. These units are responsible for carrying out these tasks simultaneously. It is not the biological parallels between these systems that are relevant; rather, it is the functional performance of these systems that is important; this is what makes them consistent with the principles of functionalist psychology.

The idea of strong artificial intelligence, which asserts that machines might in reality have mental states if they were to acquire the appropriate level of functional organization, is supported by the theory of functionalism as well. This theory adds support to the concept of strong artificial intelligence. An artificial intelligence system that is capable of perceiving, reasoning, learning, and acting in a manner that is functionally equivalent to human cognition is able to possess beliefs, wants, and even consciousness, according to this point of view. This is because the system is able to perceive, reason, learn, and act in a manner that is like human

cognition. Behaviourism, which reduces mentality to observable behaviour alone, and identity theory, which ties mental states entirely to biological brains, are two approaches that stand in contrast to this viewpoint. Both of these perspectives seek to explain mental states by relating them to bodily brains.

Semantic networks, conceptual graphs, frames, rule-based systems, and procedural models are some examples of the representational and computational techniques that are utilised extensively by artificial intelligence (AI), in addition to logical reasoning strategies. By providing methods that are both practical and versatile for encoding and manipulating knowledge, these approaches make it possible for machines to deal with ambiguity, context, and the processing of huge amounts of information. In this regard, artificial intelligence systems frequently outperform both the neural network (NN) framework and many Western epistemological traditions. These traditions have a tendency to place an excessive amount of stress on determining the identity of the knower or the unique location where knowledge is acquired. Despite the fact that such difficulties are significant from a philosophical standpoint, it is possible that they are of minimal value to advanced artificial intelligence systems, which are able to function efficiently without a unified, conscious subject of knowledge.

Western epistemology was responsible for the development of a significant number of the conceptual tools that enabled the development of artificial intelligence. Structure, reasoning, and the evaluation of belief have been areas of emphasis throughout the past, beginning with Greek logic and continuing into current conceptions of justification. It is possible to view artificial intelligence as an extension of this endeavour, which is the endeavour to comprehend and recreate knowledge using formal means. On the other hand, ongoing discussions about meaning, consciousness, and uncertainty serve to remind us that the philosophical concerns are certainly not yet resolved.

Knock your Brain

1. Which pramāṇa is closest to AI sensor data?

A. Perception

B. Testimony

C. Postulation

D. Comparison

2. Learning from patterns in data in AI is similar to:

A. Yoga

B. Anumāna (inference)

C. Worship

D. Myth

3. Trusting information from a database in AI resembles:

A. Illusion

B. Śabda (testimony)

C. Doubt

D. Meditation

4 A major concern in both Indian epistemology and AI is:

A. Fashion

B. Validity of knowledge

C. Architecture

D. Music

5. Nyāya philosophers analyzed error in order to:

A. . Improve valid cognition

B Increase confusion

C. Reject perception

D. Avoid debate

Unleash your potential

1. Examine the theory of knowing (pramāṇa) within Indian philosophy and its significance for artificial intelligence.

2.Examine the epistemic concepts of Nyaya and Buddhist philosophy concerning artificial intelligence.

3. Analyse Behaviourism, Identity Theory, and Functionalism concerning artificial intelligence.